Informatica®

Informatica® MDM - Customer 360
10.5 HotFix 5

Installation and Configuration Guide

Informatica MDM - Customer 360 Installation and Configuration Guide
10.5 HotFix 5
December 2025

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Preface

Follow the instruction in *Informatica® MDM - Customer 360 Installation and Configuration Guide* to install and upgrade Informatica MDM - Customer 360. The guide also includes pre-installation and post-installation tasks, pre-upgrade and post-upgrade tasks, and information to troubleshoot common issues.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Network

The Informatica Network is the gateway to many resources, including the Informatica Knowledge Base and Informatica Global Customer Support. To enter the Informatica Network, visit https://network.informatica.com.

As an Informatica Network member, you have the following options:

- Search the Knowledge Base for product resources.
- View product availability information.
- Create and review your support cases.
- Find your local Informatica User Group Network and collaborate with your peers.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

Informatica Product Availability Matrices

Product Availability Matrices (PAMs) indicate the versions of the operating systems, databases, and types of data sources and targets that a product release supports. You can browse the Informatica PAMs at https://network.informatica.com/community/informatica-network/product-availability-matrices.

Informatica Velocity

Informatica Velocity is a collection of tips and best practices developed by Informatica Professional Services and based on real-world experiences from hundreds of data management projects. Informatica Velocity represents the collective knowledge of Informatica consultants who work with organizations around the world to plan, develop, deploy, and maintain successful data management solutions.

You can find Informatica Velocity resources at http://velocity.informatica.com. If you have questions, comments, or ideas about Informatica Velocity, contact Informatica Professional Services at ips@informatica.com.

Informatica Marketplace

The Informatica Marketplace is a forum where you can find solutions that extend and enhance your Informatica implementations. Leverage any of the hundreds of solutions from Informatica developers and partners on the Marketplace to improve your productivity and speed up time to implementation on your projects. You can find the Informatica Marketplace at https://marketplace.informatica.com.

Informatica Global Customer Support

You can contact a Global Support Center by telephone or through the Informatica Network.

To find your local Informatica Global Customer Support telephone number, visit the Informatica website at the following link:
https://www.informatica.com/services-and-training/customer-success-services/contact-us.html.

To find online support resources on the Informatica Network, visit https://network.informatica.com and select the eSupport option.

CHAPTER 1

Introduction to Informatica MDM - Customer 360

This chapter includes the following topics:

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Overview of Informatica MDM - Customer 360

Informatica MDM - Customer 360 provides clean, consistent, and connected information about customers. Business managers use the master customer data to make better business decisions about customers and manage customer relationships. You can get a single trusted view of a customer.

With Customer 360, business users can achieve the following goals:

- Streamline the customer onboarding and qualification process.
- Centralize the data about customers in a master database.
- View the relationships between customers, parent companies, subsidiaries, and related organizations.
- Design campaigns based on customer preferences and behavior.
- Improve customer service.
- Connect customer data to the product catalogs and view the buying patterns.
- View customer-to-product relationships.

Customer 360 is based on Multidomain MDM. Business users connect to master customer data through a business-friendly user interface. The user interface displays an enterprise-level dashboard as well as 360 degree customer views that are customized for different business users.

The Customer 360 environment integrates with Informatica Product 360, which is a product information management system.

Use Customer 360 to access the master data that is stored in the Hub Store. Master data is organized by business entities. Business entities are the type of top-level data that has significance for an organization. Business entities include persons and organizations.

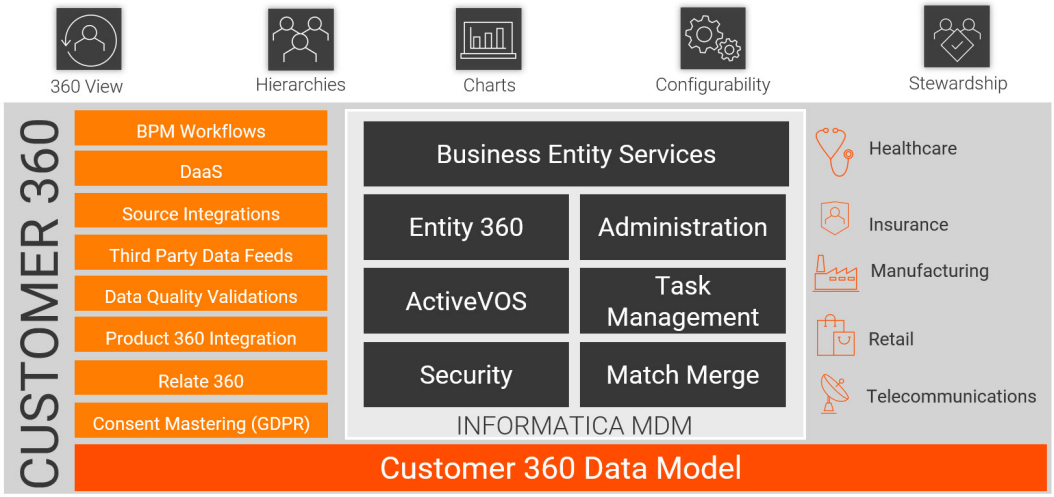
Business users use Customer 360 to perform the following activities:

- Add records.
- Search for and edit records.
- Manage relationships between records.
- Participate in review processes for new and changed records.

Architecture

Customer 360 requires Multidomain MDM and the embedded Informatica ActiveVOS® Server. For product information, you need integration with Product 360.

The following diagram shows the architecture for Customer 360:



Customer Data Management

Customer 360 adds additional features to the Informatica MDM environment, such as a database schema for customer data, business processes for customer management, and an interface that the business users can use to access customer data.

Customer data management includes onboarding of a customer and customer profile management.

Informatica Multidomain MDM

Customer 360 includes the standard Multidomain MDM components:

Hub Store

Databases that store and consolidate business data. The Multidomain MDM consists of an MDM Hub Master Database and Operational Reference Stores. The application ships with a database schema that you use to create an Operational Reference Store for customer master data.

Hub Server

A J2EE application that you deploy on an application server. The Hub Server processes data within the Hub Store and integrates the MDM Hub with external applications. The Multidomain MDM is the run-time

component that manages core and common services for the MDM Hub. It also manages user authentication across all components.

Process Server

A J2EE application that you deploy on an application server. The Process Server cleanses and matches data and performs batch jobs such as load, recalculates best version of the truth, and revalidates. The Process Server interfaces with cleanse engines to standardize the data and to optimize the data for match and consolidation.

ActiveVOS Server

Business process management software that automates business processes. The application ships with business processes that help you to manage the customer lifecycle. These processes ensure that authorized business managers review customer profiles and review internal updates to master data.

Data Director

A browser-based interface that business managers use to view and manage data. The Application ships with Customer 360, which contains an enterprise-level workspace focused on customers and views designed for business managers.

Hub Console

A browser-based interface that administrators use to manage the MDM Hub and data stewards use for managing records and batch processing of records.

Informatica MDM - Product 360

Informatica MDM - Product 360 creates a single repository for all product data. In Customer 360, MDM - Product 360 is an optional component.

Product 360 supplies the product data and the purchase history of a customer. You can view the data in the customer dashboard.

Extensions

The Customer 360 extensions extend the Customer 360 data model to address specific use cases.

For example, the consent mastering extension stores information related to customer consent to comply with General Data Protection Regulation (GDPR) for protection and privacy of customer data. Similarly, the Insurance 360 and Healthcare 360 extensions store information that are specific to insurance and healthcare domains.

To use a Customer 360 extension, install the extension.

Consent Mastering

The Customer 360 Consent Mastering extension is designed to store consent data to comply with data regulatory body such as GDPR.

The Consent Mastering extension enables customers to opt in or flag the processing of personal data. Use the Consent Mastering extension to store the consents of the customers to process the personal data such as name, address, email address, and phone number. You can also store the consent proofs, customer privacy policies, purpose of data processing, and information about the organizations that process the data.

When you install Customer 360, you get an option to install the consent mastering extension.

Insurance 360

The Insurance 360 extension is an extension that's designed for the insurance domain. Use the Insurance 360 extension to create master data from multiple insurance policy and claim management systems. You can use the master data to improve data governance and compliance, efficiently manage life events of customers, and identify potential upselling and cross-selling opportunities.

You can also extend the Insurance 360-related base objects to manage additional business needs. For example, you can extend the agreement base object to store risk, producer, employment, and account agreements.

When you install Customer 360, you get an option to install the Insurance 360 extension.

The Insurance 360 extension includes the following predefined business entities:

- Auto Policy
- Land Vehicle
- Life Policy
- Organization Policy
- Policy
- Product
- Person Policy
- Property Policy
- Structure

Healthcare 360

The Healthcare 360 extension caters to the needs of the healthcare domain. Use the Healthcare 360 extension to manage healthcare data, such as the records of patients, healthcare organizations, healthcare professionals, health plans, payments, and supporting staff members. You can create master data from multiple healthcare management systems to create a holistic view of the healthcare data.

You can also use master data to assist healthcare payers and providers identify consumer behaviors across the healthcare ecosystem and improve customer engagement and loyalty. For example, a healthcare payer organization can uniquely identify the relationships between the members and classify them as subscribers or dependents to improve customer engagement. You can also extend the Healthcare 360-related based objects to manage additional business needs.

When you install Customer 360, you get an option to install the Healthcare 360 extension.

The Healthcare 360 extension includes the following predefined business entities:

- Address
- Contract
- Health Care Organization
- Health Care Provider
- Member
- Patient
- Payer - Individual
- Payer - Organization
- Plan

- Staff

User Roles

The MDM Hub user roles control read and write privileges of the Operational Reference Store (ORS) that contains the customer master data.

Customer 360 uses the following MDM Hub user roles:

User roles for system users

The ApplicationAdministrator role is for a super user, who has full privileges.

User roles for business users

User roles for the business users control the data privileges in the ORS and also the review privileges in the business processes. Each business user who is authorized to participate in customer data management receives one or more role assignments. Many users can have the same role.

Customer 360 includes the following predefined roles for the business users:

- Sales Operation Analyst
- Sales Operation Manager
- Task Administrator

Data as a Service Validation

Use the integration of Customer 360 with Informatica DaaS for data verification and correction. You can verify customer contact information and store accurate data.

You can analyze, verify, correct, and format addresses according to the local postal standards. You can ensure that the email addresses are valid and formatted correctly. You can find email domains involved with spam networks, traps, and other malicious threats and verify that the email address is valid, not valid, or malicious. You can verify the phone numbers of your customers.

CHAPTER 2

Before You Install

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Read the Release Notes

Read the *MDM - Customer 360 Release Notes*. You can also find information about the known limitations for the release.

Software Requirements

Set up the Informatica Multidomain MDM environment before you install Informatica MDM - Customer 360.

Perform the following tasks:

1. Review the Product Availability Matrix for Customer 360 to discover the system requirements and supported versions for products, databases, and application servers. You can find all Product Availability Matrices at https://network.informatica.com/community/informatica-network/product-availability-matrices.
2. Install a supported version of Multidomain MDM and the embedded version of Informatica ActiveVOS® Server.
3. If you plan to use Informatica MDM - Product 360, install a supported version.

Extracting the Customer 360 Application Archive File

You receive the Customer 360 application as an archive file. Create the following directory structure and extract the contents of the Customer 360 archive file into it:

```
<MDM Installation Directory>/app/tcr
```

The extracted content contains the following files and folders:

File or Folder Name	Description
batchgroup	Contains the JAR file for the silent installation process.
bin/	Contains installation, upgrade, and database schema validation utilities.
bpm/	Contains the ActiveVOS default business process in a deployable format. The folder contains the following sub-folders: - pre_c360_10.2_hf2/. Directory that contains the default business processes for the data model based on the Party Role table. - source/. Contains the source files for business processes. - InfaCustomer360Workflow.bpr. Contains the default business process for the data model based on the Party table.
config/	Contains configuration properties files.
CustomerB2BPortal	Contains the files of the preconfigured Customer Portal for Organization business entity.
data/	Contains the scripts for inserting lookup or reference data.
datamodelmigration/	Contains the scripts and the change lists required to migrate from the data model based on the Party Role table to the data model based on the Party table.

File or Folder Name	Description
docs/	Contains the data dictionary. The folder contains the following files and sub-folders: - C360_104HF1_DataDictionary_en.pdf. Data dictionary for the data model based on the Party table. - pre_c360_10.2_hf2. Directory that contains the data dictionary for the data model based on the Party Role table.
email-config	Contains the subdirectories that contain configuration files for customer portal email configuration. Following are the list of the subdirectories: - templates/. Contains the avos-templates with email body text templates for ActiveVOS. - emailConfig.xml. File containing the configuration properties for email templates.
extensions	Contains the files for Customer 360 extensions.
externalcall	Contains the folders and files for SOAP services.
hub/	Contains the subdirectories that contain the database schema and the configuration files to deploy. Following are the list of the subdirectories: - change-xml/. Contains the MDM Hub metadata including components, such as landing tables, lookup tables, staging tables, base objects, and match and merge rules, cleanse functions, component instances, business entities, and business entity services. - cocs-config-xml/. Contains configuration files for the business entities and business entity services. - delta_change_xml/. Contains the newly added MDM Hub metadata. - entity360config-xml/. Contains copies of the Entity 360 component instance definitions that ship with Multidomain MDM.
idd/	Contains the message and error bundle files.
lib/	Directory for the external libraries. Copy the JDBC driver files for your database to the lib directory.
localizationScript	Contains the scripts for localizing labels and error messages.
lookuplocalization/	Contains the localized lookup data.
notices/	Contains the legal copyright file.
reporting/	Contains datamart folder which contains the data mart service and the chart configurations. The pre_c360_10.2_hf2/ directory in the datamart folder contains the data mart service for the data model based on the Party Role table.
resources/	Contains the resource bundle.properties files for each of the supported locales.
upgrade/	Contains the supporting files for the upgrade utility.
user_exit/	Contains the MDM Hub user exit archive.

File or Folder Name	Description
was	Contains the Provisioning tool user interface for a WebSphere environment.
bundleLocalization.jar	JAR file for localization.
Customer360ExternalCall.ear	File for SOAP service to validate the customer data.
domain-validation.jar	JAR file for validating Customer 360 domain.
email-config-util	JAR file for email configuration.
MDMAppsServices.war	File that contains Customer 360 services.
mdmappsview-ear.ear and uiwebapp-ear.ear	Files that contain Customer 360 user interface JBoss.
productversion.jar	JAR file for the product version.
provisioning-ear.ear	File that contains the Provisioning tool user interface.
store-credentials.jar	File that contains credential details.

Copying Compatible JAR File

To successfully upgrade and ensure compatibility with latest JDK, complete the following steps:

1. Navigate to the following directory:

   `<MDM installation directory>/app/tcr/`
2. Copy the `domain-validation.jar` file and copy to the following directory:

   `<MDM installation directory>/hub/server/lib/`
3. Restart the application server.
4. Validate the Supplier 360 Operational Reference Store (ORS) from the Hub Console.

   If the domain validation passes successfully, proceed with the upgrade of Supplier 360 components.

Creating the Operational Reference Store

Create an Operational Reference Store (ORS) for Customer 360.

For more information about how to create an ORS, see the *Multidomain MDM Installation Guide*.

Importing the Metadata

After you create the Customer 360 Operational Reference Store, import the metadata into the Operational Reference Store.

For more information about how to import the metadata, see the *Multidomain MDM Installation Guide*.

Registering the Operational Reference Store

Register the Customer 360 Operational Reference Store.

1. Log in to the MDM Hub Console.
2. In the Configuration workbench, click **Databases**.
3. On the **Write Lock** menu, click **Acquire Lock**.
4. Click **Register database**.

   The **Informatica MDM Hub Connection Wizard** appears.
5. Follow the online instructions to specify the same parameters that you specified when you created the Operational Reference Store.
6. On the **Summary** page, click **Test Database**.

   The Wizard tests the database connection parameters.
7. Ensure that you select **Create datasource after registration**.
8. Click **Finish**.
9. When prompted, enter the user credentials for the application server.

   The wizard creates a data source.

Importing the Application Metadata from a Change List

The metadata for the Customer 360 database schema resides in an MDM Hub change list. You import the change list into the Hub Store. The metadata change list creates components, such as landing tables, user exits, lookup tables, staging tables, base objects, and match and merge rules. The created tables are empty.

Before you import the application metadata, you must start the Hierarchy Manager in the MDM Hub Console. For more information about starting the Hierarchy Manager, see the *Multidomain MDM Data Steward Guide*.

1. In the **Configuration** workbench of the Hub Console, click **Repository Manager**.
2. On the **Import** tab, click the button next to the **Source** field.

   The **Open Repository** dialog box appears.
3. On the **File Repository** tab, browse to the following directory:

   `<MDM Installation Directory>/app/tcr/hub/change-xml`
4. Select the `TCR_HUB.change.xml` file, and click **OK**.

5. In the **Target** list, select the Customer 360 Operational Reference Store.
6. Select all the schema components, and click **Apply**.

   The Repository Manager imports the selected components from the change list to the Customer 360 Operational Reference Store.

Configuring the Lookup Scripts for Silent Installation

If you plan to install Customer 360 in silent mode, update the lookup script based on the database you use.

1. Go the following directory:

   `<MDM installation directory>/app/tcr/data/reference_data/`
2. If you use Oracle database, perform the following steps:
   a. In a text editor, open the `C360_lookup_script_oracle.sql` file.
   b. Comment the following line of code:

      ```
      SET DEFINE OFF
      ```
3. If you use Microsoft SQL Server database, perform the following steps:
   a. In a text editor, open the `C360_lookup_script_MSSQL.sql` file.
   b. Comment the following line of code:

      ```
      BEGIN TRANSACTION
      ```
4. Save the file.

Inserting Reference Data for Customer 360

After you import the metadata, you can populate the tables with some reference data. By using reference data, you can complete the configuration steps faster because you do not have to manually insert the rows into the tables.

Before you insert the reference data, you must configure a Process Server. The Process Server is a servlet that cleanses data and processes batch jobs. For more information about configuring a Process Server, see the *Multidomain MDM Configuration Guide*.

1. Open a command prompt, and navigate to the following directory:

   `<MDM Installation Directory>/app/tcr/data/reference-data`
2. Run one of the following scripts based on the database you use:
   - For Oracle. `C360_lookup_script_oracle.sql`
   - For Microsoft SQL Server. `C360_lookup_script_MSSQL.sql`
   - For IBM DB2. `C360_lookup_script_DB2.sql`
3. To verify whether the lookup records are loaded successfully, perform the following tasks:
   a. In the Hub Console, in the Utilities workbench, click **Batch Group**.
   b. Expand **BG_All_Lookup_Load**, and select **Control & Logs**.

c. In the Logs for each job table, review the **Status** column to verify that the load is successful. The **Total records** column shows the number of records added. The columns to the right of the **Total records** column displays zeros if all the records load successfully.

d. If the load is unsuccessful, try running the load. Select **BG_All_Lookup_Load** and click **Execute**.

Importing and Deploying the Customer 360 Application

Import the `Customer360.zip` file as an IDD application, and then deploy it.

1. To start Data Director Configuration Manager, perform the following tasks:
   a. Open a supported web browser.
   b. Enter the following URL:

      `https://<Home name:<Port number>/bdd/config`

   c. Enter the user name and password, and then click **Log In**.

   The Data Director Configuration Manager starts and the applications page appears.
2. Click **Import** > **Import complete IDD application (Zip)**.
3. Click **Browse**, and navigate to the following directory:

   `/app/tcr/idd/config`
4. Select `Customer360.zip`, and click **Open**.
5. Click **Import**.
6. Under **Bind Logical ORS**, select Operational Reference Store (ORS) from the list.

   You must select the ORS that you created for Customer 360 when you installed Multidomain MDM.
7. Click **Save**.

   Data Director imports and validates the IDD application. The process might take some time.
8. Ignore any validation errors.
9. With the application selected, click **Application State** > **Full Deployment**.

   Data Director deploys the IDD application.

Informatica Data as a Service

To validate and verify postal addresses, email addresses, and phone numbers, use the Informatica Data as a Service (DaaS) cleanse functions. To use a validation service, add the mandatory parameters for the service in the MDM Hub.

Each service contains default values for the optional parameters. If required, change the default values in the Hub Console.

Use the following rules for the validation service parameters:

- Address validation. Prepend the parameter names with `ADV6_`.
  By default, `ADV6` uses REST API. To use ADV5, add the following parameter and value:

  ```
  OPERATION_TYPE: ADV5
  ```

  The ADV5 doesn't support address standardization, output fields, and enrichments.
- Email address verification. Prepend the parameter names with `EMV6_`.
- Phone number validation. Prepend the parameter names with `GPV15_`.

The following table lists the mandatory parameters for the DaaS validation services:

Service	Mandatory Parameter
Address validation	- `LOGIN` - `PASSWORD` - `END_POINT` for ADV5 - `REST_ENDPOINT` for ADV6 **Note:** When https is enabled, you must add or override the value for the `ENDPOINT` parameter.
Country type	`COUNTRYTYPE`. Set the parameter value to ISO2.
Email address verification	`LICENSE_KEY`
Phone number validation	`LICENSE_KEY`

Specifying the Mandatory Parameters for DaaS Validation

Configure the mandatory parameters for DaaS validation in the Hub Console.

1. Log in to the MDM Hub Console.
2. Select the Customer 360 ORS.
3. In the Model workbench, click **Cleanse Functions**.

   The Cleanse Functions tool appears.
4. On the **Write Lock** menu, click **Acquire Lock**.
5. Click **Informatica Data as a Service**.
6. Click the **Add** button.
7. Specify the parameter name and value.
8. Repeat steps 6 and 7 to add the other mandatory parameters.
9. Save the changes, and click **Refresh**.

Configuring Address Standardization

You can use Address Verification to define general standardization for the elements in an address, such as locality, country, building, postal code, street, house number, and post office.

You can standardize addresses at the following levels:

Address Level

To standardize complete addresses, use the address level standardization properties, such as `Casing`, `maxItemLength`, `Alias Handling`, `CountryCodeType`, `DescriptorLength`, `FormatWithCountry`, and `CountryNameLanguage`.

For example, to modify the casing of the complete address, append the `casing` property with the `ADV6_Standardizations_Default` default standardization parameter. Set the parameter value to `Lower`, `Mixed`, or `Upper`.

You can refer to the following format:

```
ADV6_Standardizations_Default_Casing=Lower
```

Element Level

To standardize discrete elements in addresses, use the element level standardization properties, such as `casing`, `maxItemCount`, `maxItemLength`, and `Alias Handling`.

For example, the existing address line 1 appears in four lines. To display line 1 of addresses in a single line, append the `MaxItemCount` property with the `ADV6_Standardizations_ElementStandardizations_AddressLine1` default standardization parameter. Set the parameter value to `1`.

You can refer to the following format:

```
ADV6_Standardizations_ElementStandardizations_AddressLine1_MaxItemCount=1
```

For more information about the different address-level and element-level properties for configuring the standardization parameters, see the *Informatica Address Verification Developer Guide*.

Note: To validate and generate correct output data, ensure that you add valid properties as specified in Address Verification. If you add invalid properties, Address Verification generates extraneous output data.

Configuring Output Details

When you process an address, the output contains the list of fields, such as address elements, address line elements, enrichment values, and preformatted elements.

For more information about different output properties in Address Verification, see the *Informatica Address Verification Developer Guide*.

Configuring Single Address Line Element

You can configure properties to specify how Address Verification interprets addresses and generates output for addresses that you enter in a single line.

For example, if you want addresses to appear in single line with delimiters, you can append properties, such as `SingleAddressLine` and `SingleAddressLineDelimiter` with the `ADV6_OutputDetail` default parameter.

For the `SingleAddressLine` and `SingleAddressLineDelimiter` properties, you can view the following values:

Name	Value
`ADV6_OutputDetail_PreformattedData_SingleAddressLine`	`true`
`ADV6_OutputDetail_PreformattedData_SingleAddressLineDelimiter`	`Semicolon`

For more information about the different single address line elements in Address Verification, see the *Informatica Address Verification Developer Guide*.

Enabling Geocode

To use the Address Verification service with geocode, cameo, country specific, and certification enrichments, use the Hub Console to enable the enrichments.

1. Log in to the Hub Console.
2. Select the Customer 360 ORS.
3. In the Model workbench, click **Cleanse Functions**.

   The Cleanse Functions tool appears.
4. On the **Write Lock** menu, click **Acquire Lock**.
5. Expand **Informatica Data as a Service** and select **Address Verification**.
6. On the **Test** tab, enter the values for the following cleanse functions:

Input Name	Value
`EnableGeocode`	Y
`EnableCertificateMode`	Y
`EnableCountrySpecific`	Y
`validate`	Y

7. Save the changes, and click **Refresh**.
8. Click **Lock Release** > **Write Lock**.

Configuring Geocode, Country Specific, and Certification Enrichments

Address Verification can enrich addresses with additional information. Enrichments include information about regions to which addresses belong and help the postal service to find the destination mailboxes. You can configure enrichments, such as `Geocode`, `Country Specific`, and `Certification`.

To configure enrichments, use the following default parameter formats:

Enrichment	Default Parameter Format
Geocode	`ADV6_Enrichments_Geocoding_<Geocoordinates>`
Country Specific	`ADV6_CountrySpecific_<Country code>_<Output response key>`
Certification	`ADV6_Certifications_<Certification type>_<Output response key>`

Geocode Enrichment

You can use the Geocoding element to add geocoordinates as an enrichment to a verified address. Geocoordinates indicate the latitude and the longitude of an address.

You can configure different properties using the geocode enrichments. For example, if you want to configure the `Rooftop` property, you must configure the name and value in the following format:

```
ADV6_Enrichments_Geocoding_Rooftop=true
```

Country Specific Enrichment

Informatica Address Verification provides additional information as enrichments to addresses from several countries.

For example, the USA country has a country specific output value as `CountyFipsCode`. If you want to map the country with `CountrySpecific Output Line 1` , you must configure the name and value in the following format:

```
ADV6_CountrySpecific_USA_CountyFipsCode=CountrySpecific Output Line 1
```

Certification Enrichment

Postal certification improves the quality of addresses and ensures that Address Verification services meet postal authority requirements. Certifications indicate if an address contains the data required by the certification standards of national mail carriers.

For example, the CASS Certifications Type has the Certifications Type output value as `ErrorCode`. If you want map the certification type with `Certified Output Line 1`, you must configure the name and value in the following format:

```
ADV6_Certifications_CASS_ErrorCode=Certified Output Line 1
```

For more information about the different enrichments and their properties, see the *Informatica Address Verification Developer Guide*.

Creating a JCEKS Keystore

Create a JCEKS keystore to store the Informatica MDM - Product 360 credentials.

1. Run the following command:

   ```
   <JAVA_HOME>\bin\keytool.exe" -genseckey -alias <aliasname_for_keystore> -keyalg
   <Encryption Algorithm> -keysize <KEY_SIZE> -storetype jceks -keystore
   <name_of_keystore>.keystore
   ```

   Note: Recommended encryption algorithm is AES and key size is 256.

   The following sample command uses AES encryption algorithm and key size of 256:

   ```
   C:\Program Files (x86)\Java\jdk1.8.0_91\bin\keytool.exe" -genseckey -alias test -keyalg
   AES -keysize 256 -storetype jceks -keystore test.keystore
   ```

2. When prompted, provide a keystore password.

   Note: For future use, note the password.

3. When prompted for key password, press **Enter**.

4. To confirm that the key is generated, run the following command:

   ```
   JAVA_HOME\bin\keytool.exe" -list -v -storetype jceks -keystore
   <name_of_keystore>.keystore
   ```

   The first entry with the given alias name appears.

Creating an ApplicationAdministrator User

The ApplicationAdministrator role is a role with all the privileges.

1. Create an MDM Hub user.
2. Assign the ApplicationAdministrator role to the user.

Creating a User in the MDM Hub

Create an MDM Hub user before you assign a role to the user.

1. In the Hub Console, in the Configuration workbench, click **Users**.
2. Acquire a write lock.
3. On the **Users** tab, click the **Add user** button.

   The **Add User** dialog box appears.
4. Enter the first, middle, and last names for the user.
5. Enter the user name for the user. The user must use the user name logs in to the Hub Console.
6. Enter the default database for the user, which is the Operational Reference Store that contains the customer master data.
7. Enter and verify a password for the user.
8. Click **OK**.

Assigning the ApplicationAdministrator Role

After you create the user, assign the ApplicationAdministrator role to a user.

1. In the Hub Console, connect to the Operational Reference Store for Customer 360.
2. Acquire a write lock.
3. In the Security Access Manager workbench, click **Users and Groups**.

   The Users and Groups tool opens. You can use the role-first approach or user-first approach.
4. If you want to follow the role-first approach, on the **Assign Users/Groups to Role** tab, perform the following steps:
   a. Select the **ApplicationAdministrator** role.
   b. Click the **Edit** button.

      The **Assign Users to Role** dialog box appears.
   c. Select the user that you created earlier.
   d. Click **OK**.
5. If you want to follow the user-first approach, on the **Assign Roles to User/Group** tab, perform the following steps:
   a. Select the user you created earlier.
   b. Click the **Edit** button.

      The **Assign Users to Role** dialog box appears.
   c. Select **ApplicationAdministrator**.

d. Click **OK**.

Configuring the Application Properties File

In the `application.properties` file, configure the properties related to file import, JMS queue, and ActiveVOS.

1. Navigate to the following directory:

   `<MDM installation directory>/app/tcr/config`

2. Open the `application.properties` file in an editor, and configure the following properties:

   activevos.datasource.url

   Specifies the JNDI name of the ActiveVOS data source.

3. Save the file.

Configuring the Keystore Properties

Specify the properties related to the keystore in the `mdmapps-config.properties` file.

1. Navigate to the following directory:

   `<MDM installation directory>/app/tcr/config`

2. Open the `mdmapps-config.properties` file in an editor.
3. Specify the following properties:

Property	Description
keystore.file.path	Path to keystore.
keystore.pass.property.path	Path to the keystore password file, which is `keystore-pass.properties`.
application.admin.user	Name of the ApplicationAdministrator user that you created earlier.
base.url	Base URL for the business entity services. For example, `http://<Host>:<Port>`

4. Save the file.

Configuring the Keystore Password

Specify the keystore password in the `keystore-pass.properties` file.

1. Navigate to the following directory:

   `<MDM installation directory>/app/tcr/config`
2. Open the `keystore-pass.properties` file in an editor.
3. Specify the `keystore.password` parameter.
4. Save the file.

Configuring the Properties File for Silent Installation

If you want to install the Customer 360 application without user interaction in silent mode, configure the `C360_silent_installer.properties` file. When you perform the silent installation, the installer reads the `C360_silent_installer.properties` file to determine the installation options. Ensure that you provide correct settings in the properties file.

1. Go to the following directory:

   `<MDM installation directory>/app/tcr/config`
2. Open the `C360_silent_installer.properties` file.
3. Set the values for the required parameters in the `C360_silent_installer.properties` file in a text editor.
4. To validate and upgrade multiple Customer 360 Operational Reference Stores, add the list of Operational Reference Stores in the `ORS_ID` property.

   For example, `ORS_ID=Schema1,Schema2`
5. To access the Customer 360 Operational Reference Stores, add the following properties for each Operational Reference Store:

Property	Description
`<ORS ID 1>.USERNAME`	User name to access the Operational Reference Store 1.
`<ORS ID 1>.PASSWORD`	Password for the user name.

6. Comment all the properties added for upgrade.
7. Save the file.

Configuring the Log File Path

Specify the path to the log file in the `mdmapps-log4j.properties` file.

1. Navigate to the following directory:

   `<MDM installation directory>/app/tcr/config`
2. Open the `mdmapps-log4j.properties` file in an editor.
3. Specify the following log file properties:

Property	Description
`appender.file.fileName`	Path to the log file. For example, `appender.file.fileName=<MDM installation directory>/mdmapplogs/mdmapps.log` **Note:** If you plan to install Supplier 360 in the same environment, ensure that you specify a location that is external to both the applications. A common file stores the logs for both the applications.
`appender.rolling.filePattern`	Pattern for the log file name. For example, `appender.rolling.filePattern=<MDM installation directory>/mdmapplogs/mdmapps-%i.log` You can refer to the following format of the log file name: `<MDM installation directory>/mdmapplogs/mdmapps-1.log`

4. Save the file.

Editing the Resource Bundle Files

The resource bundle files contain the locale specific validation messages. If required, you can edit the files to add custom validation messages.

1. Navigate to the following directory:

   `/app/tcr/resource`
2. Open the locale specific file that you want to edit.
3. Edit the file as required.
4. Save the file.

Copying the JDBC Drivers

Before you install Customer 360, copy the JDBC driver for your database to the following directory: `/app/tcr/lib`

Configuring the Portal URL

If you plan to use Customer Portal or a custom portal, update the value of the `portal.cmx.url` property in the `cmxserver.properties` file.

1. Go to the following directory:

   `<MDM installation directory>/hub/server/resources`
2. In a text editor, open the `cmxserver.properties` file.
3. For the `portal.cmx.url` property, update the value with the host name and port number of the portal in the following format:
   - Secure connections. `portal.cmx.url=https://<MDM Hub Server host name>:<MDM Hub Server port number>`
   - Non-secure connections. `portal.cmx.url=http://<MDM Hub Server host name>:<MDM Hub Server port number>`
4. Save the file.

Configuring Security Headers for the Portal

If you want to enable all the security headers in the Customer Portal using the content security policy, add the `portal.security.httpHeaders.enabled` and `portal.security.httpHeaders.Content-Security-Policy` properties in the `cmxserver.properties` file. The content security policy is a layer of security that protects the portal from external attacks.

1. Navigate to the following directory:

   `<MDM installation directory>/hub/server/resources`
2. In a text editor, open the `cmxserver.properties` file.
3. To enable security headers in the portal URL and configure the content security policy, add the following properties:

Property	Value
`portal.security.httpHeaders.enabled`	`true`
`portal.security.httpHeaders.Content-Security-Policy`	`default-src 'self'; frame-ancestors 'self';script-src 'self'`

You can configure the `portal.security.httpHeaders.Content-Security-Policy` property only if you set the `portal.security.httpHeaders.enabled` property to true.

4. Save the `cmxserver.properties` file.

CHAPTER 3

Installing MDM - Customer 360

This chapter includes the following topics:

- Installation Overview, 31
- Installing MDM - Customer 360 in Console Mode, 31
- Installing MDM - Customer 360 in Silent Mode, 32
- Install the Portal Configuration Tool, 33

Installation Overview

You can install the Customer 360 application in console or silent mode.

When you install Customer 360 in console mode, you need manual interaction to provide the required installation options during the process of installation. When you install Customer 360 in silent mode, you do not need any manual interaction during the process of installation.

Installing MDM - Customer 360 in Console Mode

After you perform the pre-installation tasks, you can install MDM - Customer 360.

1. Go to the following directory:

   `<MDM installation directory>/app/tcr/bin`
2. Run one of the following scripts:
   - On Windows. `install-tcr.bat`
   - On Linux. `install-tcr.sh`

3. At the prompts, enter the following parameters:

Parameter	Description
MDM Hub installation directory	Fully qualified path to the directory where you installed MDM Hub.
MDM - Customer 360 installation directory	Fully qualified path to the directory that contains the Customer 360 files.
Application Server	Name of the application server in lowercase. For example, weblogic, jboss, websphere.
avos console username	User name with administrative privileges to access the ActiveVOS Console.
avos console password	Password of the ActiveVOS Console user name.
PIM portal username	Product 360 user name. If you do not want to enable Product 360, press **Enter**.
PIM portal password	Product 360 password.

Note: If you use a Weblogic application server, you are prompted to enter the Weblogic console password. After you successfully install the Customer 360 application, you can install the Customer 360 extensions that you want.

Installing MDM - Customer 360 in Silent Mode

You can install Customer 360 in silent mode without any user interaction. Before you install Customer 360 in silent mode, ensure that you configure the `C360_silent_installer.properties` file.

1. Open a command prompt, and navigate to the following directory:

   `<MDM installation directory>/app/tcr/bin`

2. Run the following command:

   - On Windows. `install-tcr.bat silent <MDM installation directory>\app\tcr\config\C360_silent_installer.properties`
   - On UNIX. `./install-tcr.sh silent <MDM installation directory>/app/tcr/config/C360_silent_installer.properties`

   Note: The installer runs in the background. The process can take a while to complete. After the installation is complete, review the messages to ensure the successful installation of Customer 360.

3. After you install Customer 360, download and apply EBF-21763 and EBF-21766.

   Note: For more information on applying emergency bug fixes (EBF), contact Informatica Global Customer Support.

Install the Portal Configuration Tool

Use the Portal Configuration tool to import and customize the preconfigured Customer Portal or create a custom portal.

1. At a command prompt, navigate to the following directory:

   `<MDM installation directory>/app/portal/bin`

2. Run one of the following scripts:

   - On Windows. `install-portal.bat`
   - On Linux. `./install-portal.sh`

3. At the prompts, enter the following parameters:

Parameter	Description
MDM Hub installation directory	Press Enter to use the default path or type the fully-qualified path to the directory where you installed the MDM Hub.
Portal Configuration tool installation directory	Press Enter to use the default path or type the fully qualified path to the directory where you plan to install the Portal Configuration tool.
Application server	Type the name of the application server in lowercase. Use one of the following values: - weblogic - jboss - websphere
URL of the portal	Enter the URL to access the portal in the following format: - Secure connections. `https://<MDM Hub Server host name>:<MDM Server port number>/` - Non-secure connections. `http://<MDM Hub Server host name>:<MDM Server port number>/`
MDM Hub administrator user name	Type the user name with administrative privileges to access the MDM Hub.
ActiveVOS Console user name	Type the user name with administrative privileges to access the ActiveVOS Console.
Password for the ActiveVOS Console user name	Type the password of the ActiveVOS Console user name.

The script installs the Portal Configuration tool. After you install the Portal Configuration tool, you can import the preconfigured Customer Portal.

CHAPTER 4

After You Install

This chapter includes the following topics:

- Installing Customer 360 Extensions, 34
- Adding User Accounts and Assign Roles, 35
- Configure the Operational Reference Store, 37
- Populate Customer 360 Charts with Data, 38
- Configuring Default Hierarchy for the Customer Dashboard, 41
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- Improving Performance of Bulk Data Import, 42
- Configuring the Product 360 Connection Parameters, 44
- Integrating Informatica MDM - Relate 360, 45
- Test Customer 360, 48
- Importing the Customer Portal, 49
- Configuring the SOAP Service, 49
- Import the Security Certificate and Keystore for the Portal Configuration Tool, 50
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Installing Customer 360 Extensions

After you install Customer 360, you can install Customer 360 extensions.

You can choose to install the following Customer 360 extensions:

- Consent Management
- Insurance 360

- Healthcare 360

1. Go to the directory that contains the extension you want to install.

   The Consent Management extension is in the following directory:

   `<MDM installation directory>/app/tcr/extensions/GDPR/bin`

   The Insurance 360 extension is in the following directory:

   `<MDM installation directory>/app/tcr/extensions/INFAIN/bin`

   The Healthcare 360 extension is in the following directory:

   `<MDM installation directory>/app/tcr/extensions/INFAHC/bin`

2. Run one of the following scripts:
   - On Windows. `install.bat`
   - On Linux. `install.sh`
3. At the prompts, enter the following parameters:

Parameter	Description
MDM Hub installation directory	Fully qualified path to the directory where you installed MDM Hub.
MDM - Customer 360 installation directory	Fully qualified path to the directory that contains the Customer 360 files.
Application Server	Name of the application server in lowercase.
avos console username	ActiveVOS Console user name.
avos console password	ActiveVOS Console password.

Note: If you use a Weblogic application server, you are prompted to enter the Weblogic console password.

Adding User Accounts and Assign Roles

User roles for the business managers control the data privileges in the MDM Hub and review privileges in the business processes. Each data steward or business user who is authorized to participate in customer data management receives one or more role assignments. Many users can have the same role.

1. If the business users who need to use the Customer 360 do not have the MDM Hub user accounts, add a user account for each business user.
2. Assign roles to the users.

For more information about users and roles, see the *Multidomain MDM Security Guide*.

Adding User Accounts

If some business users do not have the MDM Hub user accounts, create the user accounts.

Before you begin, you can review an existing MDM Hub user account to see which authentication type is used in the Informatica MDM environment.

1. In the Hub Console, in the Configuration workbench, click **Users**.
2. Acquire a write lock.
3. On the **Users** tab, click the **Add user** button.

   The **Add User** dialog box appears.
4. Enter the first, middle, and last names for the user.
5. Enter the user name for the user. The user must use the user name to log in to the Hub Console.
6. Enter the default database for the user, which is the Operational Reference Store that contains the customer master data.
7. Enter and verify a password for the user.
8. Choose the type of authentication.
   - If your MDM Hub implementation uses authentication through a third-party security provider, select **Use external authentication**.
   - If you want to use the internal authentication in the MDM Hub, clear **Use external authentication**.
9. Click **OK**.

   The Users tool adds the user to the list of users.

Role Privileges

The following table lists the user roles and their privileges:

Role	Add or edit a customer profile	Review and Approve customer profiles or profile updates
ApplicationAdministrator	Yes	Yes
Sales Operation Analyst	Yes	No
Sales Operation Manager	Yes	Yes

Assigning Roles to Business Users

You need to assign user roles to the business users who are responsible for reviewing the new customer profiles and the edited customer profiles.

Use the Hub Console to assign user roles. You can follow the role-first or user-first approach. For a role-first approach, you select a role and then select the users and user groups that you want to associate with the role. For a user-first approach, you select a user or user group and then select the roles for the user or user group.

1. In the Hub Console, connect to the Operational Reference Store of Customer 360.
2. Acquire a write lock.
3. In the Security Access Manager workbench, click **Users and Groups**.

The Users and Groups tool opens.

4. If you want to follow the role-first approach, on the **Assign Users/Groups to Role** tab, perform the following steps:
   a. Select a workflow role.
   b. Click the **Edit** button.

      The **Assign users to Role** dialog box appears.
   c. Select the users and user groups to whom you want to assign the role.
   d. Click **OK**.
   e. Repeat the steps for other workflow roles.
5. If you want to follow the user-first approach, on the **Assign Roles to User/Group** tab, perform the following steps:
   a. Select a user or user group.
   b. Click the **Edit** button.

      The **Assign users to Role** dialog box appears.
   c. Select the workflow roles suitable for the user or user group.
   d. Click **OK**.
   e. Repeat the steps for other users and user groups.

Configure the Operational Reference Store

Before you start the Customer 360 application, configure the Operational Reference Store that contains customer data. You must disable a trigger, truncate the data in a repository table, and review the value of the GETLIST Limit property.

Truncating a Repository Table

If you imported the Oracle database dump, the repository table C_REPOS_RPT_DETAILS contains sample data. You must truncate the data.

1. In a database tool, connect to the Operational Reference Store for C360.
2. Run the following command:

```
truncate table C_REPOS_RPT_DETAILS
```

Populate Customer 360 Charts with Data

The Home page of Customer 360 displays charts with metrics or data about customers and tasks. All charts are available as chart components except the Customer Distribution By Geography chart which is available as an external link. The data for the chart that is available as external link comes from a data mart.

The data mart service retrieves data from the Operational Reference Store and stores the results in a repository table. When the Home page loads, the page queries the data mart service for the data and populates the charts.

Note: In the data mart service and configuration files, the term *report* refers to the chart configuration.

To populate the charts, perform the following tasks:

1. Import the chart configurations into a repository table.
2. Configure a database connection between the data mart and the database that contains the Operational Reference Store.
3. Configure the report parameters to include the database name.
4. Populate the data mart with report data.

Importing Customer 360 Chart Configurations

To import the chart configurations, run an insert script on the Customer 360 Operational Reference Store (ORS). The script imports the chart configurations into the C_REPOS_RPT_CONFIG repository table.

1. Open a command prompt.
2. Navigate to the following directory:

   `<MDM installation directory>/app/tcr/reporting/datamart/chart-config`
3. Use a database tool to run the `insert_c_repos_rpt_config.sql` script.

   For example, log in with the ORS user name and password and the service name and then run the script.

   ```
   .../chart-config> sqlplus <C360 ORS>/password@service
   SQL> @insert_c_repos_rpt_config.sql
   ```

   The script inserts the charts into the C_REPOS_RPT_CONFIG repository table.

Configuring the Data Mart Database Connection

Before you can generate reports or populate charts, you must configure the data mart database connection.

1. Navigate to the following directory:

   `<MDM Installation Directory>/app/tcr/reporting/datamart/lib`
2. If the directory is empty, copy the following files from the `<MDM installation directory>/hub/server/lib` directory to the `<MDM installation directory>/app/tcr/reporting/datamart/lib` directory:
   - `log4j-api-2.17.2.jar`
   - `log4j-core-2.17.2.jar`
   - `log4j-1.2-api-2.17.2.jar`
   - `ojdbc8.jar lib/db2jcc.jar`

- db2jcc_license_cu.jar
- siperian-common.jar
- commons-validator-<version number>.jar
- jakarta.xml.bind-api-4.0.2.jar
- jaxb-runtime-4.0.2.jar
- jaxb-core-4.0.2.jar
- istack-commons-runtime-4.2.0.jar
- jakarta.activation-api-2.1.3.jar

3. Navigate to the following directory:

   <MDM installation directory>/app/tcr/reporting/datamart

4. Extract the contents of the populate_datamart.jar file.
5. Open the following file in an editor:

   <Directory containing extracted files>/META-INF/MANIFEST.MF

6. To the Class-Path parameter, add the following value:

   lib/commons-validator-<version number>.jar

7. Save the file.
8. Open a command prompt.
9. Run the following command:

   ```
   java -jar populate_datamart.jar config
   ```

10. At the prompt, type **C** to configure the database connection.
11. Answer the prompts described in the following table:

Prompt	Description
Connection Name	Enter a unique name for the connection. If the name exists, the connection is overwritten.
Type of Connection	Enter the type of connection to the data mart. Currently only DB is supported.
Database vendor	Enter the database to connect with, such as Oracle, IBM DB2, or Microsoft SQL Server.
Connection Type	Enter the one of the following connection type: Service or SID.
User	Enter the database user.
Password	Enter the database password.
Token	Reserved for future use.
Host Name	Enter the database host name.
Port	Enter the database port.
Database Name	Enter the database name/SID.

12. When prompted to finish the configuration, enter **Y**.

    The connection information is saved in the following file: `<MDM installation directory>/app/tcr/reporting/datamart/config/mart-population-config.xml`

Configuring the Chart Parameters

Before you can populate the data mart, you must configure the parameters related to the charts.

1. Open a command prompt.
2. Navigate to the data mart directory.

   `<MDM installation directory>/app/tcr/reporting/datamart`
3. Run the following command:

   ```
   java -jar populate_datamart.jar config
   ```
4. Type **P** to configure the report parameters.
5. Enter appropriate values for the chart.

   The following table lists the parameters to configure for the chart:

Prompt	Description
Report Name	Name of the chart as specified in the following file: `/app/tcr/reporting/datamart/config/report-class-mapping.properties`
Report Configuration ID	Unique chart identifier as specified in the C_REPOS_RPT_CONFIG repository table.
Mart Connection Name	Connection name for the data mart to connect with the Operational Reference Store (ORS). Use the connection name defined for the ORS that contains the customer data.
Query Connection Name	Connection name for the database to be queried. - For reports that include the word Tasks, specify the connection name defined for the ActiveVOS database. - For all other reports, specify the connection name defined for the ORS that contains the customer data.

6. When prompted to finish the configuration, enter **N**.
7. Repeat the steps for other charts.
8. After you enter all tables, exit the configuration tool.

   The parameters are saved in the following file: `<MDM installation directory>/app/tcr/reporting/datamart/config/mart-population-config.xml`.

Populating the Data Mart with Data

Run a `java` command to populate the data mart with data for all charts or for a specific chart. If you want to specify a chart, use its report name.

1. Open a command prompt.
2. To populate the data mart with data for all the available reports, run the following command:

   ```
   java -jar populate_datamart.jar exec
   ```

3. To populate the data mart with data for a specific report, run the following command:

   ```
   java -jar populate_datamart.jar exec <report name>
   ```

   Important: You cannot use the service name to populate the data mart with data for a specific report.

If the MDM Hub contains data, the C_REPOS_RPT_DETAILS repository table is populated with report data. The ROWID_RPT_CONFIG column links the data to the report configuration that requested the data.

Configuring Default Hierarchy for the Customer Dashboard

In the customer dashboard, the Hierarchies panel displays how a record is related to other records in a hierarchical format. A record can belong to multiple hierarchies. If you define a default hierarchy for the hierarchy component, the Hierarchies panel loads the default hierarchy for the record. Otherwise, the Hierarchies panel loads the list of hierarchies to which the record belongs.

1. In the Provisioning tool, click **Configuration** > **Component Editor**.

   The **Component Editor** page appears.
2. To configure the organization hierarchy, select **Hierarchy Widget** > **CustomerOrgHierarchyWidget**, and then set the default hierarchy.
3. To configure the person hierarchy, select **Hierarchy Widget** > **CustomerPersonHierarchyWidget**, and then set the default hierarchy.
4. Click **Apply**.
5. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Importing the Localized Lookup Data

Customer 360 installation files include the localized lookup data. If you use a localized environment, import the localized lookup data into the database before you use Customer 360.

1. Copy the following files from the `<MDM installation directory>/hub/server/lib` directory to the `<MDM installation directory>/app/tcr/lookuplocalization/lib` directory:
   - `log4j-core-2.17.2.jar`
   - `log4j-1.2-api-2.17.2.jar`
   - `log4j-api-2.17.2.jar`

- For Oracle. `ojdbc8.jar`
- For IBM DB2. `db2jcc.jar`
- For Microsoft SQL Server. `sqljdbc4.jar`

2. Go to the following directory:

   `<MDM installation directory>/app/tcr/lookuplocalization/bin`

3. Run the following command:
   - On UNIX. `./lookup_localization.sh`
   - On Windows. `lookup_localization.bat`
4. At the prompts, enter the following parameters:

Parameter	Description
Database type	Type of database that you use. Use one of the following values: - Oracle - DB2 - MSSQL
User name	User name to access the Operational Reference Store database.
Password	Password for the user name.
Operational Reference Store database host name	Name of the host that runs the Operational Reference Store database.
Operational Reference Store database port number	Port number that the database listener uses.
Database name	For IBM DB2 and Oracle only. Name of the IBM DB2 database or Oracle service.

   The localized lookup data is imported into the staging tables.
5. In the Utilities workbench of the Hub Console, click **Batch Group**.
6. Expand **Localized Lookup Data Load**, and select **Control & Logs**.
7. Click **Execute**.

   The localized lookup data is imported into the base objects.

Improving Performance of Bulk Data Import

When you import a file in a workflow-enabled environment, the workflow triggers the approval process in parallel. The parallel processing of workflow might affect the performance of the import process when you import more than 5000 records with child records. To improve the performance of the import process in a workflow-enabled environment, you can now set a wait time for the workflow process to trigger. Based on the wait time, the workflow process gets triggered and avoids parallel processing.

1. Go to the following directory:

   `<MDM installation directory>/app/tcr/config/`

2. In a text editor, open the `c360-portal-workflow-config.xml` file.
3. Update the following properties and save the file:

Property	Description
`<property name="portal.fileImport.wait">P0Y0M0DT2H0M0S</property>`	Optional. Wait time for the workflow to trigger after the import process starts. For example, if the wait time is 2 hours 30 minutes, then the workflow triggers after 2 hours 30 minutes after the import process starts. You can specify the wait time in the following format: `P<n>Y<n>M<n>DT<n>H<n>M<n>S` - P - Indicates portal. - Y - Indicates number of years. - M - Indicates number of months. - D - Indicates number of days. - T - Indicates time. - H - Number of hours. - M - Number of minutes. - S - Number of seconds. For example, the value `P0Y0M0DT2H30M0S` indicates 2 hours and 30 minutes. Ensure that you set the `portal.fileImport` property to `Y` for the wait time to be considered. Default is 2 hours. **Note:** Informatica recommends to use 1 hour 30 minutes for 5000 records and 2 hours 30 minutes for 10000 records.
`<property name"portal.fileImport">Y</property>`	Indicates whether to enable wait time for the workflow to trigger. Set to `Y` to enable the wait time. The wait time improves the performance of import process when you import more than 5000 records. Use this option during the file import or initial data load process. After the data load is complete, change the value from `Y` to `N`. Default is `N`.

4. After you make changes to the properties, run one of the following redeploy scripts to view the changes:
   - On UNIX. `redeploy.sh`
   - On Windows. `redeploy.bat`

   Note: Alternatively, you can log in to the ActiveVOS console, navigate to the **Catalog** > **Resources** > **Other** section to update the properties in the `c360-portal-workflow-config.xml` file.
5. Go to the following directory:

   `<MDM installation directory>/hub/server/resources`
6. In a text editor, open the `cmxserver.properties` file.

7. Update the following properties and save the file:

Property	Description
`cmx.server.be-import.task-limit`	Specifies the maximum number of records users can import for the task approval workflow to trigger. Set `cmx.server.be-import.task-limit=10000` for users to import up to 10000 records and the task approval workflow to trigger. If a user attempts to import more than 10000 records, the task approval workflow does not trigger and displays an error.
`cmx.server.find-replace.task-limit`	Specifies the maximum number of replaced records that trigger the task approval workflow. Set `cmx.server.find-replace.task-limit=1000` for users to replace up to 1000 records and the task approval workflow to trigger. If a user attempts to import more than 1000 records, the task approval workflow does not trigger and displays an error.

8. Restart the application server.

Configuring the Product 360 Connection Parameters

Configure the Product 360 server connection parameters to display the product data from the party role product table in the dashboard view of a customer.

1. In the Provisioning tool, click **Configuration** > **Component Editor**.

   The **Component Editor** appears.
2. In the **Components** list, select **C360 Product Widget**, and then select **ProductComponent**.
3. In the **Properties** panel, specify the values in the XML code for the following parameters:

Parameter Name	Description
productHost	Name of the Product 360 server.
productPort	Port on which the host listens.

4. Optionally, edit the `structureSystem` parameter value to remove the default Heiler Standard structure system, and specify the new structure system.
5. Optionally, edit other default parameter values in the code.

   Note: The `contentType` parameter does not support the TIFF image format.
6. Click **Apply**.
7. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.

- To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Integrating Informatica MDM - Relate 360

You can integrate Relate 360 with Customer 360 to view the relationship graph. A relationship graph displays all the related records of a Person record and its relationships with the related records.

You must create a custom view to display the relationship graph for a Person record. You can also create a graph component to display an aggregate view of the related records in the Person dashboard.

To integrate Relate 360 with Customer 360, perform the following tasks:

1. Create an external resource for the relationship graph.
2. Design a layout for the Person business entity view to display the relationship graph.
3. Create an external resource for the aggregate view of the relationship graph.
4. Edit the layout of Person dashboard to include the aggregate view of the relationship graph.

Before you integrate Relate 360 with Customer 360, use Relate 360 to process the Customer 360 data and other data related to the customers.

For more information about Relate 360, see the Informatica MDM - Relate 360 documentation set.

Creating an External Resource for the Relationship Graph

An external resource is a custom component that you can add to a layout. For the relationship graph, configure an external resource that you can add to the Person business entity view.

For more information about external resource, see the *Multidomain MDM Provisioning Tool Guide*.

1. In the Provisioning tool, click **Configuration** > **Component Editor**.

   The **Component Editor** appears.
2. In the **Components** list, select **External Resources**, and click **Create**.

   The **Properties** panel appears.
3. Specify a name for the external resource.
4. Add the following XML code format for the external resource:

   ```
   <iframe src="http://<Host>:<Port>/<Relationship Graph UI WAR Name>/noshell/graph/
   <Source>/Customer/{{rowidObject}}/network" width="100%" height="800px"></iframe>
   ```

   The XML code format uses the following parameters:

   Host

   Host name or IP address of the machine on which you deploy the Relate 360 relationship graph user interface WAR file.

   Port

   Port on which the host listens.

   Relationship Graph UI WAR Name

   Name of the relationship graph user interface WAR file that you deploy on the host.

Source

Source name of the records. If you do not have the source name for the records, use the source name that you used in Relate 360 to process the customer data.

The following sample XML code creates an external resource for the relationship graph:

```
<iframe src="http://10.12.128.71:8080/bdrm-ui/noshell/graph/MDM/Customer/
{{rowidObject}}/network" width="100%" height="800px"></iframe>
```

5. Click **Apply**.
6. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Design a Layout for the Person Business Entity View

After you create an external resource for the relationship graph, design a layout to include the external resource for the Person business entity view .

1. In the Provisioning tool, click **Configuration** > **Layout Designer**.

   The **Layout Designer** appears.
2. Click **Create** > **Entity View**.

   The **Define Layout Properties** page appears.
3. Define properties for the layout that you want to design.
   a. Specify the following properties:

Property	Description
View Name	Label for the option.
Layout Name	Label for the layout that you design. After you design and publish the layout, the layout name appears in the **Layout Categories** panel.
Layout ID	Label used for generating a system ID.
Description	Optional. A meaningful description to identify the layout.

   b. To assign the layout to the Person business entity view, select **Person** > **CustomPersonView**.
4. Click **Next**.

   The **Define Purpose** page appears.
5. Specify the purpose of the layout that you want to design.
   a. Select the user roles that can access the layout.

      The user roles that appear on the **Define Purpose** page are configured in the MDM Hub environment.
   b. Select **Edit entities**.

6. Click **Next**.

   The **Select a Template** page appears.

7. Select **Template 1**, and click **Next**.

   The **Design** page appears.

8. Drag the external resource that you created for the relationship graph into the workspace.

9. Click **Save**.

   The changes are saved to the temporary workspace.

10. Publish the changes to the MDM Hub.

    a. Click **Publish**.

       A confirmation dialog box appears that prompts you to publish or review the changes.

    b. Review the changes or publish without a review.

       - To publish without a review, click **Publish**.
       - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Creating an External Resource for the Aggregate View

Configure an external resource for the aggregate view of the relationship graph. The aggregate view displays an aggregated view of the related records for a Person business entity.

1. In the Provisioning tool, click **Configuration** > **Component Editor**.

   The **Component Editor** appears.

2. In the **Components** list, select **External Resources**, and click **Create**.

   The **Properties** panel appears.

3. Specify a name for the external resource.

4. Add the following XML code format for the external resource:

```
<iframe src="http://<Host>:<Port>/<Relationship Graph UI WAR Name>/noshell/graph/
<Source>/Customer/{{rowidObject}}/aggregate" width="100%" height="800px"></iframe>
```

   The XML code format uses the following parameters:

   Host

   Host name or IP address of the machine on which you deploy the Relate 360 relationship graph user interface WAR file.

   Port

   Port on which the host listens.

   Relationship Graph UI WAR Name

   Name of the relationship graph user interface WAR file that you deploy on the host.

   Source

   Source name of the records. If you do not have the source name for the records, use the source name that you used in Relate 360 to process the customer data.

   The following sample XML code creates an external resource for the aggregate view:

```
<iframe src="http://10.12.128.71:8080/bdrm-ui/noshell/graph/MDM/Customer/
{{rowidObject}}/aggregate" width="100%" height="800px"></iframe>
```

5. Click **Apply**.

6. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Edit the Layout of Person Dashboard

After you create an external source for the aggregate view of the relationship graph, edit the layout of Person dashboard to include the aggregate view.

1. In the Provisioning tool, click **Configuration** > **Layout Designer**.

   The **Layout Designer** appears.
2. Expand the **CustomerPerson** layout category, and select **layout2**.
3. Click **Edit**.

   The **Design** page appears.
4. Drag the external resource that you created for the aggregate view into the workspace.
5. Click **Save**.

   The changes are saved to the temporary workspace.
6. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Test Customer 360

You can log in to Customer 360 with your administrative user credentials.

1. Use the following URL format to access Customer 360:

   ```
   http://<MDM Server host name>:<MDM Server port number>/mdmapps/
   com.informatica.tools.mdm.web.auth/login
   ```
2. Log in with your user credentials.
3. If prompted to select an application, select **Customer 360**.

   The **Home** page appears.

Importing the Customer Portal

If you install the Portal Configuration tool, import the preconfigured Customer Portal. Use the Portal Configuration tool to import the Customer Portal.

1. Open a supported browser.
2. Enter the URL for the Portal Configuration tool.

   The URL has the following format:

   - Secure connections. `https://<MDM Hub Server host name>:<MDM Server port number>/portal-config/`
   - Non-secure connections. `http://<MDM Hub Server host name>:<MDM Server port number>/portal-config/`

   The **Log In** page opens.
3. Enter your user name and password.
4. Click **Log In**.

   The Portal Configuration tool opens and displays the **Home** page.
5. Click **Import Portal**.

   The **Import Portal** dialog box appears.
6. Navigate to the following directory:

   `<MDM Installation Directory>/app/tcr/CustomerB2BPortal`
7. Select the `CustomerB2BPortalConfig` file.
8. Select the Operational Reference Store for the portal to use.
9. Select the source system for the portal to use.
10. Enter a unique name for the portal.
11. Click **Import**.

    The selected portal is imported.

Configuring the SOAP Service

Configure the SOAP service to validate the customer data when a business user adds or updates a customer record.

1. Log in to the Provisioning tool, and select the Customer 360 Operational Reference Store.
2. Click **Business Entity** > **Extensions**.
3. From the **Extensions** list, select **SOAP Services**, and then select **C360ExternalCallService**.
4. In the **Endpoint** field, specify the host name and port number of your application server in the following format:

   ```
   http://<host name of the application server>:<port number of the application server>/Customer360ExternalCall/CustomLogicService
   ```
5. Click **Apply**.

6. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Import the Security Certificate and Keystore for the Portal Configuration Tool

If you use a hosted MDM environment with Portal Configuration tool installed in multiple nodes and connected to a single database, ensure that the security certificate and keystore remain same in all the nodes. After you install the Portal Configuration tool in all the nodes, import the security certificate and keystore from the first node to all other nodes.

The following procedure explains steps to import the files for an environment with two nodes:

1. Copy the following file:

   ```
   <MDM installation directory in Node 1>/hub/server/resources/certificates/
   certificate_infaPortal.cert
   ```
2. Paste the copied file in the following directory:

   ```
   <MDM installation directory in Node 2>/hub/server/resources/certificates/
   ```
3. To delete the expired or old certificates from the keystore, open a command prompt and run the following command:

   ```
   keytool -delete -noprompt -alias infaportal -keystore /<MDM installation directory
   in Node 2>/hub/server/resources/certificates/MDM_KEYSTORE_FILE_JKS.keystore
   ```

   When prompted, enter the password for the keystore.
4. Copy the following file:

   ```
   <MDM installation directory in Node 1>/hub/server/resources/certificates/
   MDM_KEYSTORE_FILE_JKS.keystore
   ```
5. Paste the copied file to the following directory:

   ```
   <MDM installation directory in Node 2>/hub/
   ```
6. To import the keystore, on a command prompt, run the following command:

   ```
   keytool -v -importkeystore -srckeystore <MDM installation directory in Node 2>/hub/ -
   srcalias infaPortal -destkeystore /<MDM installation directory in Node 2>/hub/server/
   resources/certificates/MDM_KEYSTORE_FILE_JKS.keystore -deststoretype JKS
   ```

   When prompted, enter the password for the keystore.

Configuring the Content Security Policy to View the Dashboard

Configure the content security policy in the Provisioning tool to view data on the dashboard that you see on the Home page.

1. Log in to the Provisioning tool, and select the Customer 360 Operational Reference Store.
2. Click **Configuration** > **General Settings**, and then select **Content Security Policy**.
3. Click **Content Security Policy**.

   The tree view panel and the content security policy panel display the content security policy options.
4. In the **Content Security Policy** panel, click **Add Content Security Policy Tag**.
5. From the **Policy Tag** list, select **script-src-elem**.
6. Enter the following required values for the policy tag:
   - `'self'`
   - `'unsafe-inline'`
   - `*`

   Use the **Add** button to add multiple values for a policy tag. Use the **-** button to delete values.
7. To apply the updated content security policy to your system, click **Apply**.
8. Click **Publish**.
9. Review the changes and click **Confirm**.

CHAPTER 5

Business Processes for Customer Management

This chapter includes the following topics:

- Workflows, 52
- User Roles, 52
- Role Privileges, 53
- Task Types Associated with Records, 53
- Task Types Associated with Hierarchies, 54

Workflows

Business processes automate some common customer lifecycle management workflows.

Customer 360 ships with ActiveVOS business processes for the following workflows:

- Create a customer person or customer organization.
- Update a customer profile.

User Roles

User roles control the write privileges. User roles for business users control the data privileges in the ORS and also the review privileges in business processes. Each business user who is authorized to participate in reference data management receives one or more role assignments. Many users can have the same role.

Customer 360 has the following user roles:

- Sales Operation Analyst
- Sales Operation Manager
- ApplicationAdministrator

Role Privileges

The following table lists the user roles and summarizes the role privileges:

Role	Add or edit a customer person or customer organization	Approve customer person or customer organization or profile updates
ApplicationAdministrator	Yes	Yes
Sales Operation Analyst	Yes	No
Sales Operation Manager	Yes	Yes

Task Types Associated with Records

By default, workflows are triggered when a user with the Sales Operation Analyst role creates or updates a record. When a user with the Sales Operation Manager role creates or updates a record, no workflows are triggered.

The Customer 360 default configuration includes the following tasks:

Predefined Tasks	Description
DraftState	Triggered when a creator creates or updates a record and chooses to keep the change in draft state.
Notification	Triggered when an approver approves or rejects a record.
Update	Triggered when an approver sends back a record for resubmission.
FinalReview	Triggered when a creator sends a new record or an updated record for approval.

The following steps summarize what happens when a user with the Sales Operation Analyst role creates or updates a record and a user with the Sales Operation Manager role approves the record:

Task	Task Owner	Possible Actions	Result
DraftState	Creator	Submit	Record is submitted for approval. Record is assigned to the users with the Sales Operation Manager role.
		Discard	Record is deleted.
FinalReview	Approver	Claim	User claims the tasks. Record appears in the task inbox of the user.
		Disclaim	Record is assigned back to the users with the Sales Operation Manager role.

Task	Task Owner	Possible Actions	Result
		Approve	Record is promoted. Notification sent to the creator.
		Reject	Record is deleted. Notification is sent to the creator.
		Send Back	Record is sent back to the creator with comments.
Update	Creator	Submit	Record is resubmitted for approval. Moves to the FinalReview workflow.
		Discard	Record is deleted.
Notification	Creator	Ok	Closes the workflow.

Task Types Associated with Hierarchies

When a Sales Operation Manager updates the hierarchy of a record or organization, no workflow is triggered. The changes made are directly saved. When a Sales Operation Analyst makes changes related to the hierarchy of a record or organization, a one-step approval workflow is triggered.

The Customer 360 default configuration for hierarchy workflow includes the following tasks:

Task Type	Description
FinalReview	Created when a user make changes to the hierarchy of a record and sends the changes for approval.
UpdateReview	Created when an approver sends back the changes made to the hierarchy of a record for resubmission.
Notification	Created when an approver approves or rejects the changes made to the hierarchy of a record.

The following steps summarize what happens when a user with the Sales Operation Analyst role updates a hierarchy and a user with the Sales Operation Manager role approves or rejects the changes made:

Task	Task Owner	Possible Actions	Result
FinalReview	Approver	Claim	User claims the tasks. Hierarchy change appears in the task inbox of the user.
		Disclaim	The hierarchy change is assigned back to users with the Sales Operation Manager role.
		Approve	The hierarchy change is promoted. Notification is sent to the requestor.
		Reject	The hierarchy change is deleted. Notification is sent to the requestor.

Task	Task Owner	Possible Actions	Result
		Send Back	The hierarchy change is sent back to the requestor with comments.
UpdateReview	Requestor	Submit for Approval	The hierarchy change is resubmitted for approval. Moves to the FinalReview task.
		Cancel	The hierarchy change is deleted.
Notification	Requestor	OK	The workflow is closed.

CHAPTER 6

Customizing Customer 360

This chapter includes the following topics:

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- Customizable Resources and Components, 56
- Customizing the Layout, 57
- Customizing a Predefined Chart, 64
- Extending the Data Model, 65
- Extending the Customer 360 Resources, 66
- Extending the ActiveVOS Business Process, 67
- Extending the User Exit, 67
- Customizing Customer 360 Workflows for Portal Users, 68
- Localizing Customer 360, 70
- Customizing the Product 360 Component, 75

Customizing Customer 360 Overview

After you configure Customer 360, you can customize some of the features to suit your environment. The customization includes extending the data model, modifying the layout for the dashboard view of a customer, modifying the Home page, and adding custom charts and user interface components.

Customizable Resources and Components

You can customize the Customer 360 components and resources based on your specific business needs.

You can perform the following customizations:

- Customize the Customer 360 user interface and dashboard components.
- Customize the predefined charts on the Customer 360 home page.
- Add columns to base object tables, lookup tables, and relationship tables.
- Remove fields from the user interface layout for an existing business entity.
- Modify the properties of a field. For example, enable a field as searchable, required, read only, or sortable.

- Add a business entity.
- Add or modify business entity views.
- Add or modify hierarchy and network relationships.

  Note: You cannot delete an existing hierarchy or network relationship.
- Modify triggers, tasks, and templates.

  Note: You cannot delete an existing trigger, task, or template.

Note: After you upgrade, you can restore the customization from the `c_repos_co_cs_config` and `c_repos_component_instance` repository tables.

Customizing the Layout

The Customer 360 user interface consists of multiple components. Each component displays data related to a record. You can configure some of the components that appear in a Customer 360 view.

You can customize the following three user interfaces:

Home page

The default page you see when you log in to Customer 360.

Person dashboard

When you open a customer profile, you see the customer dashboard. If the customer is a person, the page you see is the Person dashboard.

Organization dashboard

If the customer is an organization, the page you see is the Organization dashboard.

The default Customer 360 layout comes with preconfigured components. The configuration is present in the `C_REPOS_COMPONENT_INSTANCE` table.

User Interface Components

The Customer 360 user interface consists of default components and external link components. You can edit the default configuration of Customer 360 user interface components.

You can edit the configuration of the following components:

- Task Manager
- My Records
- Data View

You can edit the configuration of the following external link components:

- Social
- Overview
- Financials
- Identifiers
- Indicators
- Twitter

You can also add custom external link components and charts.

Components and Parent Elements

The following table lists the parent XML elements for each of the default components:

Element	Description
components	Top level element that contains the component configuration. The element contains `taskManager`, `myRecordsManager`, `bulkImport`, and `dataView`.
taskManager	Name of the component for Task Manager. Contains the configuration for the Task Manager. The element contains `taskTypes`.
myRecordsManager	Name of the component for My Records. Contains the configuration for My Records. The element contains `taskTypes`.

Customizing the Dashboard Components

To customize the dashboard components, use the Application Editor of the Provisioning tool.

1. In the Provisioning tool, click **Configuration** > **Application Editor**.

   The **Application Editor** appears.
2. In the **Applications** list, select **MDMApps Application Configuration**, and then select **MDMAppsWidgetConfiguration**.
3. In the **Properties** panel, update the values in the XML code as required.

   For example, the following sample XML code adds a text field to the dashboard:

   ```
   <field>
      <label translate="true">C360_B2C_OVERVIEW_CUSTPRVNAME</label>
      <type>text</type>
      <coValue>prvsNm</coValue>
      <defaultValue>-</defaultValue>
      <filterAs />
   </field>
   ```
4. Click **Apply**.
5. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

XML code for the My Records Component Configuration

The following table lists the sub-elements of the `myRecordsManager` parent elements:

Element	Description
taskTypes	Parent element for all the task types. The element contains `taskType`.
taskType	Contains the configuration for a task type. For example, UpdateWithApproval or FinalReview. You configure the task types in `Tasks` configuration as part of the COCS configuration. The `Tasks` configuration is present in the `C_REPOS_CO_CS_CONFIG` repository table. Has the `name` attribute which specifies the name of the task.
role	Role of the user for which you configure the workflow. **Note:** You specify the configuration for a task type and a user role. The element contains `enableEditRecordDetails`, `enableAssignTo`, and `actions`.
enableAssignTo	Indicates whether the `assign to a user` option is available for the task type and user role. Set to `true` to make the option available to the user with the role and set to `false` to make the option unavailable.
actions	Set of actions that are available for each task. Each task type has a set of possible actions that you can perform in the task context. The element contains `action`.
action	An action, such as submit, approve, reassign, or reject, that you can perform on a task. Has the `name` attribute, which specifies the name of the action. The element contains `enableCommentOnAction`, `enableAssignTo`, `isCommentMandatory`, and `actionMessage`. `enableAssignTo`: Applicable to the task action that has the manual reassign set to true.
enableCommentOnAction	Indicates whether you have the option to add a comment when you perform a task action. Set to `true` to make the option available. Set to `false` to make the option unavailable. If set to true, specify `isCommentMandatory`.
isCommentMandatory	Indicates whether you must add a comment when you perform the action. If set to `true`, specify the `actionMessage` element.
actionMessage	Message to display when you perform the action.

If you want to edit the XML code for the configuration, ensure that you follow the XML example.

The following sample code shows the configuration of the My Records component.

```
<myRecordsManager>
   <!-- Tasktypes as configured in Tasks in CO CS Config -->
   <taskTypes>
      <taskType name="AVOSBeDraftState">
         <role name="SalesOperationAnalyst">
            <enableAssignTo>false</enableAssignTo>
            <!-- Action of task as configured in BPR(ActiveVOS Workflow) -->
            <actions>
               <action name="Submit">
                  <enableCommentOnAction>false</enableCommentOnAction>
               </action>
```

```
                <action name="Discard">
                   <enableCommentOnAction>true</enableCommentOnAction>
                   <isCommentMandatory>false</isCommentMandatory>
                   <actionMessage><![CDATA[All the unsaved changes to the record will be
lost]]></actionMessage>
                </action>
             </actions>
          </role>
       </taskType>
    </taskTypes>
</myRecordsManager>
```

XML Code for the File Import Component Configuration

The File Import component configuration is used to specify the templates that you can use to import data into Customer 360. Ensure that you follow the XML example to write the XML code for the configuration correctly.

The following table lists the sub-elements of File Import element:

Element	Description
templates	Top-level element. Contains: `template`
template	Name of the import template. Has the `name` attribute.

The following sample code shows the File Import component configuration:

```
<bulkImport>
   <templates>
      <template name="CustomerPersonView" />
      <template name="CustomerOrgView" />
      <template name="ContactView" />
   </templates>
</bulkImport>
```

XML Code for the Data View Component Configuration

The Data View component configuration specifies the record for which the **Draft** check box must be displayed.

The following table lists the sub-elements of the Data View element:

Element	Description
draft	Parent element. Contains: `draftCoName` and `draftDefault`
draftCoName	Record name for which the **Draft** checkbox must be displayed.
draftDefault	Indicates whether the **Draft** check box must be selected by default.

The following sample code shows the Data View component configuration:

```
<dataView>
   <draft>
```

```
        <draftCoName>draftInd</draftCoName>
        <!-- CO name of the draft fieldl which is to be rendered -->
        <draftDefault>true</draftDefault>
        <!-- Default value of the field -->
    </draft>
</dataView>
```

XML Code for the External Link Components

The `widgets` element contains the configuration for the external link components.

The following table lists the sub-elements in the `widgets` section:

Element	Description
widget	Parent element that contains the configuration for a widget or external link.
social	Contains the URLs for social media. For example, `<Facebook>http://www.facebook.com</Facebook>`. Ensure that you use the same format in the implementation configuration. The element contains `Facebook`, `Twitter`, and other social media URLs.
b2b	Contains the configuration for a business-to-business customer profile.
overview	Title of the overview section. The element contains `title` and `section`.
title	Title of a section in the user interface.
section	Contains the configuration for a column in the widget. The element contains `field`.
field	Contains the configuration for a field in the user interface. Has the following attributes: - `group`. Indicates whether the fields must be displayed as a group. - `view`. Can be set to `horizontal` to show the field labels and values in the same line. Else, the value appears below the field. The element contains `label`, `type`, `coValue`, `defaultValue`, `defaultKey`, `defaultKeyValue`, `getStrategy`, `coFieldValues`, `concatenateWith`, and `filterAs`.
label	Text that appears for the field in the user interface.
type	Specifies the data type of the field. Has the following data types: - `img`. Indicates image. - `text`. Indicates text. - `address`. Indicates address. - `phone`. Indicates phone number. - `email`. Indicates email address. - `social`. Indicates social. - `currency`. Indicates currency. - `textWithIcon`. Indicates text with an icon. Replace with CSS class.
coValue	Record child or field, including the path.

Element	Description
getStrategy	Determines how a value is selected when the field has multiple values. Use the element only for child records. Has the following values: - `defaultOrFirstOne`. Default value or the first value in the list is selected. You can either specify the default value or specify the business entity field that contains the value. When you have multiple child records, you need to specify which of the record to select. Specify the name of the field and the value in the field. - `getOne`. Get the first available value. - `all`. Get all values. - `getByEffectiveStartDate`. Effective start date. If the effective start date is not available, then the last updated date is used to select the value.
defaultValue	If the desired value is not found, the value to be displayed for the field in the user interface.
defaultKey	If you specify the default strategy as `defaultOrFirstOne`, specify the name of the business entity field. For example, default indicator.
defaultKeyValue	If you specify the default strategy as `defaultOrFirstOne`, specify the value of the default key.
coFieldValues	Contains the business entity fields that contain the value you want to display on the user interface. Follow the parent-child relationships until you reach the required element of field.
concatenateWith	Specifies how to concatenate the fields. For example, comma-separated or space.
filter	Reserved for future use.

The following sample code shows the configuration of the `identifiers` external link component:

```
<identifiers>
   <title>Identifiers</title>
   <section>
      <field>
         <label>DUNS</label>
         <type>text</type>
         <coValue>CustomerOrgRole_item_0_CustomerOrgIdentifiers</coValue>
         <defaultValue>-</defaultValue>
         <defaultKey>dfltInd</defaultKey>
         <defaultKeyValue>1</defaultKeyValue>
         <getStrategy>getByEffectiveStartDate[effStrtDt]_altIdTyp_idTyp[DUNS]</
getStrategy>
         <coFieldValues>altIdVal</coFieldValues>
         <concatenateWith />
         <filterAs />
      </field>
   </section>
   <section>
      <field>
         <label>ParentDUNS</label>
         <type>text</type>
         <coValue>CustomerOrgRole_item_0_CustomerOrgIdentifiers</coValue>
         <defaultValue>-</defaultValue>
         <defaultKey>dfltInd</defaultKey>
         <defaultKeyValue>1</defaultKeyValue>
         <getStrategy>getByEffectiveStartDate[effStrtDt]_altIdTyp_idTyp[ParentDUNS]</
getStrategy>
         <coFieldValues>altIdVal</coFieldValues>
```

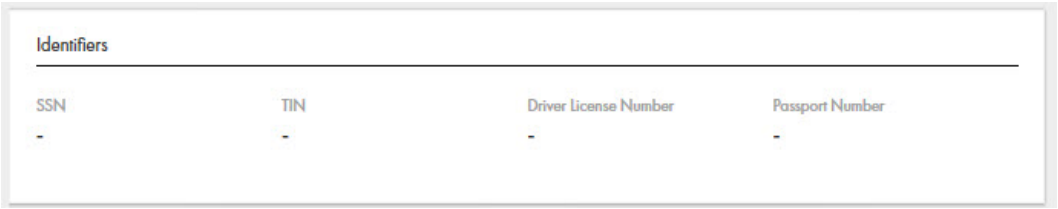
```
            <concatenateWith />
            <filterAs />
         </field>
      </section>
      <section>
         <field>
            <label>NAICS</label>
            <type>text</type>
            <coValue>CustomerOrgClassification</coValue>
            <coFieldValues>Classification_classifVal</coFieldValues>

   <getStrategy>getByEffectiveStartDate[effStrtDt]_Classification_classifTyp_classifTyp[NAIC
   S]</getStrategy>
            <defaultValue>-</defaultValue>
            <concatenateWith />
            <filterAs />
         </field>
      </section>
      <section>
         <field>
            <label>SIC</label>
            <type>text</type>
            <coValue>CustomerOrgClassification</coValue>
            <coFieldValues>Classification_classifVal</coFieldValues>

   <getStrategy>getByEffectiveStartDate[effStrtDt]_Classification_classifTyp_classifTyp[SIC]
   </getStrategy>
            <defaultValue>-</defaultValue>
            <concatenateWith />
            <filterAs />
         </field>
      </section>
   </identifiers>
```

The sample code shows the configuration of four identifiers, SSN, TIN, Driver License Number, and Passport Number.

The following image shows the Identifiers section in the user interface:



XML Code for the Twitter Component

The configurable parameter in the Twitter component configuration is the number of tweets.

The following sample code shows the Twitter configuration:

```
<twitter>
   <numberOfTweets>5</numberOfTweets>
   <section>
      <field>
         <label>Social</label>
         <type>social</type>
         <coValue>CustomerOrgRole_item_0_CustomerOrgSocialHandle</coValue>
         <defaultValue>-</defaultValue>
         <defaultKey>dfltInd</defaultKey>
         <defaultKeyValue>1</defaultKeyValue>
         <coFieldValues>sclMdaVal</coFieldValues>
         <concatenateWith>_</concatenateWith>
         <getStrategy>getOne_sclMdaTyp_sclMediaTyp[Twitter]</getStrategy>
         <filterAs />
```

```
        </field>
      </section>
    </twitter>
```

Use the `numberofTweets` element to specify the number of tweets that you want to display in the user interface. All other XML elements are standard elements that you use in an external link configuration.

Cloning the predefined layout

The predefined Customer 360 layout has preconfigured components. Ensure that you don't modify the preconfigured components. If you want to update the components, you must create a copy of the predefined layout and update the components in the cloned layout.

To customize the predefined layout, use the Application Editor of the Provisioning tool.

1. In the Provisioning tool, click **Configuration** > **Layout Designer** .

   The **Layout Designer** page appears.
2. In the **Layout Categories** section, select the predefined layout.
3. Click **Edit**.
4. In the **Data Component** list, select the predefined layout.
5. To create a copy of the predefined layout, click **Settings**.
6. On the **Record Details Component Designer** page, modify the configuration name.

   For example, if the predefined layout name is Person, you can modify the configuration name as Person_copy.
7. Click **Done**.
8. To save the layout copy, click **Save**.

You can now update the components in the cloned layout.

Customizing a Predefined Chart

The Home page displays charts that are predefined for Customer 360. All charts are available as chart components except the Customer Distribution By Geography chart which is available as an external link.

The following charts are available as chart components:

Chart	Description
Customers Added By Year	Displays the number of customers added in each year.
Customer By Onboarding Time	Displays how long it took to qualify customers.
Source Systems	Displays the percentage of customer data that comes from each source system in each year.
Tasks Status Overview	Displays the total number of tasks and breaks down the tasks by status.
Task Priority Overview	Displays the total number of tasks and breaks down the tasks by priority.
Assigned Tasks By Users	Displays the total number of tasks assigned to each user.

Chart	Description
Closed Tasks By Users	Displays the total number of tasks closed by each user.
Open Tasks By Task Type	Displays the total number of tasks that are open for each task type.
Assigned Tasks By User Roles	Displays the total number of tasks assigned to each user role.

You can customize the charts that are available as chart components to change the chart characteristics or the information the chart shows. You cannot customize the charts that are available as external links.

Use the Provisioning tool to customize a chart that is available as chart component. To customize any predefined chart, Informatica recommends that you create a chart component based on the chart that you want to customize and add it to the Home page layout.

For more information about creating a chart component, see the *Multidomain MDM Provisioning Tool Guide*.

Extending the Data Model

You can extend the Customer 360 data model by changing the physical schema or by adding types and values to some of the existing tables. You can also add new tables and attributes.

To extend the data model, perform the following steps:

1. Compare your business requirements with the existing schema.
2. List the tables and columns that you want to add.
3. Take a backup of the existing schema.
4. Review the guidelines to extend the data model.
5. Add the tables and columns.

Guidelines for Extending the Data Model

You can modify the definitions of tables or add new tables to the database.

Consider the following guidelines when you extend the data model:

- Check if you can use an existing child base object.
- Do not add a root base object to store the person or organization information.
- Do not define tables with names greater than 24 characters.
- Do not delete existing base objects.
- Do not delete the existing relationship of base objects.
- Do not delete existing columns.
- Do not modify the physical names of existing base objects. However, you can modify the display names.
- Do not modify the data type of an existing column.
- Do not decrease the length of an existing column.

- Prefix the names of the custom base objects to distinguish them from the existing tables. The prefix indicates the type of table. Use the following naming convention when you create a base object:

Prefix	Table
C_XO_	Root or child record base object.
C_XR_	Relationship base object.
C_XT_	Lookup base object.

- If you add a column to an existing table, prefix the name of a column with X_.

For more information about adding tables and columns, see the *Multidomain MDM Configuration Guide*.

Guidelines for Adding Base Objects

You can add base objects to extend the data model. You can add root or child base objects, lookup base objects, and relationship base objects.

Consider the following guidelines when you add a base object:

- **Child base object with one-to-many relationship**. Use the foreign key from the Party table to relate the table.
- **Child base object with many-to-many relationship**. Use the relationship base object to relate the table to the Party table.
- **Lookup base object**. Set the LookupIndicator to true.

Extending the Customer 360 Resources

Based on the business requirements, you can extend the Customer 360 resources. For example, to extend business entities, you can create a business entity or a lookup business entity. After you create a business entity, you can add a business entity view. You can also add a child, field, or reference fields to an existing business entity.

Guidelines for Extending the Customer 360 Resources

When you extend the Customer 360 resources, follow the naming guidelines so that you can distinguish the custom resources from the predefined Customer 360 resources.

The following table lists the naming guidelines for different Customer 360 resources:

Resource	Guidelines
Business entity and business entity view	- Add the prefix `Ex` to the names of the new business entities or lookup entities. For example, `Ex<Business Entity Name>`. - Add the prefix `Ex` to the names of the new child, field, or reference field. For example, `Ex<Business Entity Child Name>`. - Do not add a new view to an existing business entity. Create a new business entity, and then add the business entity view. - Do not delete the one-to-many or one-to-one relationship of the child node. - Do not delete a business entity field or a business entity reference field. - Do not delete a business entity or a business entity view.
Relationship type, queries, packages hierarchy, and hierarchy code	Add the prefix `X`.
Staging tables	Add the prefix `C_X`.
Business entity relationships, transformations, business entity service extensions, and component instances, such as layouts and external links	Add the prefix `Ex`.
Cleanse functions	Add the prefix `X`.
Chart components	Add the prefix `X`.

Extending the ActiveVOS Business Process

You can extend the ActiveVOS business process to include additional steps in the workflow. Customer 360 installation files include the source code to extend the ActiveVOS business process as an ActiveVOS project.

You can find the ActiveVOS project file in the following directory: `<MDM installation directory>/app/tcr/bpm/source/InfaCustomer360Workflow`

Extending the User Exit

The data model based on the Party Role table uses the post-merge user exit. You can extend the user exit to perform custom processing. Customer 360 installation files include the source code to extend the user exit as a Java project.

You can find the Java project file in the following directory: `<MDM installation directory>/app/tcr/user_exit/hub/source/CustomerUserExit`

Customizing Customer 360 Workflows for Portal Users

After you customize the data model, configure the `c360-portal-workflow-config.xml` file to reflect the data model changes for workflows.

1. Go to the following directory:

   `<MDM installation directory>/app/tcr/config/`

2. Open the following file in a text editor:

   `c360-portal-workflow-config.xml`

3. If you use customized business entity fields, update the following properties based on your requirement:

Property	Description
portal.draft.path	Business entity field that stores the value of the draft indicator for a portal user.
portal.fullname.path	Business entity field that stores the full name for the portal user.
portal.portalId.path	Business entity field that stores the portal ID for portal association.
hub.admin.user	Business entity field that stores the admin user details.
portal.SupplierRegistrationView.userInd.path	Business entity field that stores values that indicate whether a user is a portal user.

4. If you use customized values for portal user state, update the following properties:

Property	Description
portal.AVOSBeDraftState.Submit	Name of the user state when a data steward submits the user details for review.
portal.direct.Submit	Name of the user state when a business manager directly submits the user details.
portal.AVOSBeFinalReview.Send Back	Name of the user state when a business manager sends back a pending user record for additional details.
portal.AVOSBeFinalReview.Approve	Name of the user state when a business manager approves a portal user.
portal.AVOSBeUpdate.Submit For Approval	Name of the user state when a business user updates an existing portal user record and submits the changes for review.

5. If you have configure a customized password policy in the MDM Hub, update the same values for the following properties:

Property	Description
portal.passwordpolicy.maximumPasswordLength	Maximum length of the password.
portal.passwordpolicy.minimumPasswordLength	Minimum length of the password.
portal.passwordpolicy.minimumUniqueCharacters	Minimum number of unique characters in the password.
portal.passwordpolicy.patternValidationEnabled	Indicates whether the pattern validation is enabled for the password.
portal.passwordpolicy.passwordBeginningPattern	The character type allowed for the first character of the password when the pattern validation is enabled.
portal.passwordpolicy.passwordMiddlePattern	The character type allowed for the first and last characters of the password when the pattern validation is enabled.
portal.passwordpolicy.passwordEndPattern	The character type allowed for the last character of the password when the pattern validation is enabled.

6. If you use a customized email template, update the following properties:

Property	Description
portal.emailTemplate.registrationFailureC360B2BPortal	Business entity field that stores the name of the email template that is triggered when a portal user is rejected.
portal.emailTemplate.registrationSuccessC360B2BPortal	Business entity field that stores the name of the email template that is triggered when a portal user is approved.
portal.emailTemplate.internalRegistrationSuccessC360B2BPortal	Business entity field that stores the name of the email template that is triggered when a business user creates a portal user successfully.
portal.emailTemplate.updateRequestC360B2BPortal	Business entity field that stores the name of the email template that is triggered when portal user updates the user details. .

7. Save the file.
8. At a command prompt, go to the following directory:

   `<MDM installation directory>/app/tcr/bin`
9. Run one of the following scripts:
   - On UNIX. `redeploy.sh`
   - On Windows. `redeploy.bat`

Localizing Customer 360

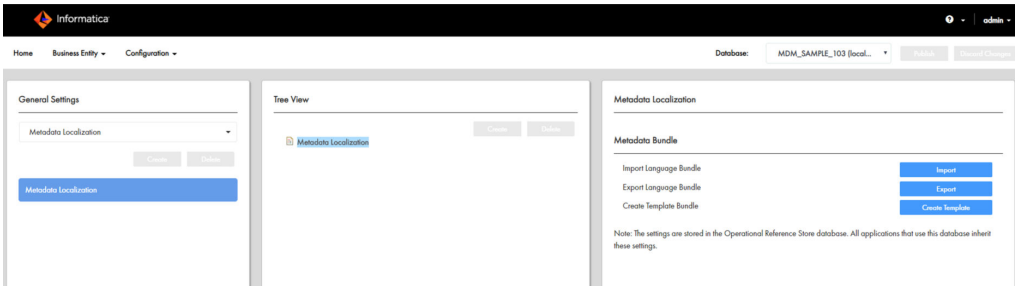
All elements of the user interface in Customer 360 are localized in the supported languages. You can localize the Customer 360 resources and custom resources in a language other than the supported languages. The customer 360 resources are menus, buttons, navigation links, labels, error messages, and metadata. The custom resources are the business entities, business entity fields, lookup tables, error messages, and labels that you add to customize Customer 360.

Localizing Metadata

The business entity metadata refers to the resources related to the data model, such as business entity names, menu names, field names, and task types.

1. Click **Configuration** > **General Settings**, and then select **Metadata Localization**.
2. Select **Metadata Localization**.
3. Click **Metadata Localization**.

   The **Tree View** and **Metadata Localization** panels display the localization options.



4. In the **Metadata Localization** panel, under **Metadata Bundle**, click **Create Template**.
5. Save the `besMetadata.zip` file to a location on your local drive.
6. Extract the `besMetadata.zip` file to your local drive.

   The properties files for the selected languages appear.
7. If you want to add a language other than the supported language, save a copy of the `besMetadata.properties` file with the corresponding language code and country code.
   For example, to create the Canadian French version, save the file as `besMetadata_fr_CA.properties`.
8. In a text editor, open the corresponding localized file.
9. For each metadata value, add the localized value in the following format:

   ```
   <Metadata value>=<Localized value>
   ```

   For example, the following property shows a French translated value for the first name:

   ```
   Person.firstName=prénom
   ```
10. Save the changes.
11. Compress the localized properties files to a .zip file named `besMetadata`.
12. In the **Metadata Localization** panel, under **Metadata Bundle**, click **Import**, and then browse to the `besMetadata.zip` file.
13. Click **Open**.

    The language metadata bundle is imported.

Localizing Task Actions, Types, and Messages

Task actions, task types, and task action messages are part of the metadata. Define the task metadata in the `besMetadata.properties` file.

1. Click **Configuration** > **General Settings**, and then select **Metadata Localization**.
2. Click **Metadata Localization**.
3. In the **Metadata Localization** panel, under **Metadata Bundle**, click **Export**. The `besMetadata.zip` template file downloads.
4. Unzip the file, and open the `besMetadata.properties` file in an editor of your choice.
5. Add the task action and task action message properties. Edit the task type properties.

   Use the following syntax for the task properties:

Property Type	Syntax	French Localization Example
Task action	`taskaction.<Task action>=<Localized task action>`	`taskaction.Cancel_Task=Annuler la tâche` **Note:** Replace white spaces in task action names with an underscore. In the example, the white space in `Cancel Task` is replaced with an underscore, `Cancel_Task`.
Task type	`tasktype.<Task type>=<Localized task type>`	`tasktype.AVOSBeMerge=Fusionner` **Note:** To know which task name to add, you must be aware of the predefined MDM workflows that are deployed to the ActiveVOS server as processes.
Task action message	`taskactionmessage.<Task action configuration>.<Task action>=<Localized message>`	`taskactionmessage.TypicalConfig.Escalate=Vous êtes sur le point de transmettre la tâche à votre responsable.`

6. Save the properties file with the appropriate language code and country code suffix.

   For example, to localize metadata in Canadian French, the language code is `fr` and the country code is `CA`. Save the file as `besMetadata_fr_CA.properties`.
7. Compress the properties file.
8. In the **Metadata Localization** panel, under **Metadata Bundle**, click **Import**. The `besMetadata.zip` template file is imported.

Localizing Lookup Table

A lookup table is a reference table that contains a list of supported values for a column in a base object table. You can localize the lookup tables in any language you want. To localize the lookup tables, create a properties file with the localized values, and then import the localized file into the database.

1. Go to the following directory:

   `<MDM Installation Directory>/app/tcr/lookuplocalization/resources`
2. Save a copy of the `lookup.properties` file with the language code and country code for the language you want to use.

For example, to create the Canadian French version, save the file as `Lookup_fr_CA.properties`.

3. In a text editor, open the localized file you create.
4. For each lookup table value, add the localized value in the following format:
   `<lookup table value>=<Localized value>`
5. Save the file.
6. Go to the following directory:
   `<MDM Installation Directory>/server/lib`
7. Copy the following files:
   - `log4j-core-2.17.2.jar`
   - `log4j-1.2-api-2.17.2.jar`
   - `log4j-api-2.17.2.jar`
   - `ojdbc8.jar`. If you use Oracle database.
   - `db2jcc.jar`. If you use IBM DB2 database.
   - `sqljdbc4.jar`. If you use Microsoft SQL Server database.
8. Paste the copied files into the following directory:
   `<MDM Installation Directory>/app/tcr/lookuplocalization/lib`
9. Open a command prompt, and go to the following directory:
   `<MDM Installation Directory>/app/tcr/lookuplocalization/bin`
10. Run the following command:
    - On UNIX. `lookup_localization.sh`
    - On Windows. `lookup_localization.bat`
11. At the prompts, enter the following parameters:

Parameter	Description
Database type	Type of database that you use. Use one of the following values: - Oracle - DB2 - MSSQL
User name	User name to access the Operational Reference Store database.
Password	Password for the user name.
Operational Reference Store database host name	Name of the host that runs the Operational Reference Store database.
Operational Reference Store database port number	Port number that the database listener uses.
Database name	For IBM DB2 and Oracle only. Name of the IBM DB2 database or Oracle service.

The localized lookup data is imported into the staging tables.

12. In the utilities workbench of the Hub Console, click **Batch Group**.

13. Expand **Localized Lookup Data Load**, and select **Control & Logs**.
14. Click **Execute**.

    The localized lookup data is imported into the base objects.

Mapping the Lookup Tables with the Localized Lookup Tables

All lookup tables support localization. To localize lookup tables, you must map the lookup table to the corresponding lookup localization table. For example, to localize a list of gender codes, map the Lookup Gender lookup table to the Lookup Gender Localized table.

1. In the Provisioning tool, click **Business Entity** > **Modelling.**

   The **Modelling** page appears.
2. Select **Reference Entities**, and then select a lookup entity reference to localize.
   For example, select **LookupGender.**
3. In the properties panel, select **C_BT_GNDR** from the **Base Object** list.
4. Select **Localization**.

   Additional database properties and values appear based on your database table setup.
   The following table contains example values for the properties related to the C_BT_GNDR_LCL localized table:

Property	Value
Base Object	C_BT_GNDR_LCL
Key Column	GNDR_CD
Country Column	CNTRY_CD
Language Column	LANG_CD
Value Column	LOC_STRNG

   Note: When you localize lookup tables, do not configure the same table column as key and value columns. The value column displays the localized value.
5. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.
   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Localizing Labels and Error Messages

Labels and error messages are static text in the Customer 360 and Provisioning tool user interface. The text cannot be localized through the Provisioning tool interface. To localize the labels and error messages, edit the Customer 360 localization bundle files.

1. Go to the following directory:

   `<MDM Installation Directory>/app/tcr`
2. Copy the following files:
   - `customer360view-ear.ear`
   - `provisioning-ear.ear`
   - `uiwebapp-ear.ear`
3. Paste the files into the following directory:

   `<MDM Installation Directory>/app/tcr/localizationScript`
4. Go to the following directory:

   `<MDM installation directory>/hub/server/lib`
5. Copy the following files:
   - `ant-contrib-1.0b3.jar`
   - `ant-launcher.jar`
   - `ant.jar`
   - `gson-2.2.4.jar`
6. Paste the files into the following directory:

   `<MDM Installation Directory>/app/tcr/localizationScript/lib`
7. Open a command prompt, and go to the following directory:

   `<MDM Installation Directory>/app/tcr/localizationScript`
8. Run the following command:

   `ant extract_all`

   The command extracts the i18n directories containing the localized bundle files for the user interface of Customer 360 and the Provisioning tool.
9. If you want to add a language other than the supported language, go to the following directories, and then save a copy of the `bundle.properties` files in the respective directories with the corresponding language code and country code:
   - `<MDM Installation Directory>/app/tcr/localizationScript/i18n/c360/com.informatica.tools.mdm.client/res/i18n` . For localizing the user interface of the Customer 360 application.
   - `<MDM Installation Directory>/app/tcr/localizationScript/i18n/uiwebapp`. For localizing the user interface of the Customer 360 components.
   - `<MDM Installation Directory>/app/tcr/localizationScript/i18n/provisioning`. For localizing the user interface of the Provisioning tool.

   For example, to create the Canadian French version, save the file as `bundle_fr_CA.properties`.
10. In a text editor, open the corresponding localized bundle files and update them.
11. From the command prompt, run the following command:

    `ant generate_all`

The command generates the JavaScript/JSON files for the bundle files.

12. To update the .ear files with the updated localization bundle files, from the command prompt, run the following command:

    `ant update_all`

13. Copy the updated .ear files from the `<MDM Installation Directory>/app/tcr/localizationScript` directory, and then paste the files into the following directory:

    `<MDM Installation Directory>/app/tcr`

14. Go to the following directory:

    `<MDM Hub installation directory>/app/tcr/bin`

15. Run the following command:

    - On Windows. `redeploy.bat`
    - On UNIX. `redeploy.sh`

16. At the prompts, enter the following parameters:

Parameter	Description
MDM Hub installation directory	Fully qualified path to the directory where you installed the MDM Hub.
MDM - Customer 360 installation directory	Fully qualified path to the directory that contains the Customer 360 files.
Application server	Name of the application server in lowercase.

The command redeploys the `customer360view-ear.ear` file with the updated bundle files.

Customizing the Product 360 Component

In the dashboard view, the Product 360 component displays the product purchase history of a customer. You can customize the fields in the product details page that opens when you click a product.

To add or remove fields in the product details page, create a custom component, and then replace it with the existing component in the layout.

1. To create a component, perform the following steps:
   a. In the Provisioning tool, go to **Configuration** > **Component Editor**.

      The **Component Editor** appears.

   b. Select **Product Component**.

   c. Based on the business entity for which you want to customize the Product 360 component, select one of the following components:

      - **ProductComponentForPerson**. For a person business entity.
      - **ProductComponentForOrg**. For an organization business entity.

   d. From the properties panel, copy the XML code of the Product 360 component.

   e. In the **Components** panel, click **Create**.

f. In the **Properties** panel, specify a name for the component, and paste the copied XML code.

g. Edit the XML code as required.
The XML code uses the following parameters:

Parameter Name	Description
complexProductField	Lists the fields that you want to concatenate based on the specified expression. For example, in the following XML sample code, PRODUCT_GROSS_WEIGHT is a complex field. The field concatenates the values of the ArticleLogistic.GrossWeight and ArticleLogistic.GrossWeightUnit fields. `<complexProductField expression="%p1 %p2" title="PRODUCT_GROSS_WEIGHT"> <productField fieldName="ArticleLogistic.GrossWeight" id="%p1"> <fieldIdentifier>countryIdentifier</fieldIdentifier> </productField> <productField displayParam="label" fieldName="ArticleLogistic.GrossWeightUnit" id="%p2"> <fieldIdentifier>countryIdentifier</fieldIdentifier> </productField> </complexProductField>`
expression	Lists the Product 360 field IDs in the order that you want to concatenate. For example, in the sample code, %p1 and %p2 are the field IDs that the expression concatenates.
id	ID for the Product 360 field that you want to concatenate. For example, in the sample code, %p1 is the ID for the ArticleLogistic.GrossWeight field.
productField	Lists the details of a field, such as the name, title, and identifier of the field. For example, in the following XML sample code, the details of the long description field for a product is listed in the productField parameter: `<productField fieldName="ArticleLang.DescriptionShort" title="PRODUCT_NAME"> <fieldIdentifier>languageIdentifier</fieldIdentifier> </productField>`
fieldIdentifier	Identifier of the field in Product 360.
fieldName	Name of the Product 360 field that you want to add to the product details page.
displayParam	Parameter of the field that you want to display.
title	Identifier for the Product 360 field in Customer 360.

h. Click **Apply**.

2. To configure the display names for the Product 360 fields in Customer 360, perform the following steps:

a. Go to the following directory:

`<MDM installation directory>/app/tcr/resources/com.informatica.tools.mdm.web.entity360view.c360.productsWidget`

b. In a text editor, open the `bundle.properties` file.

c. Add the display name of the Product 360 field in the following format:

```
<title>=<Display name>
```

For example, `PRODUCT_GROSS_WEIGHT=Gross Weight (kg).`

3. To replace the existing component with the component that you created, perform the following steps:
   a. In the Provisioning tool, go to **Configuration** > **Layout Designer**.

      The **Layout Designer** appears.

   b. Expand the **Home Page** layout category, and then select **layout1**.
   c. Click **Edit**.

      The **Design** page appears.

   d. Remove the existing Product 360 component.
   e. Drag the Product 360 component that you created into the workspace.
   f. Click **Save**.

      The changes are saved to the temporary workspace.

4. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.

   b. Review the changes or publish without a review.
      - To publish without a review, click **Publish**.
      - To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

CHAPTER 7

Upgrading Customer 360

This chapter includes the following topics:

- Upgrade Overview, 78
- Before You Upgrade, 79
- Upgrading Customer 360 in Console Mode, 91
- Upgrading Customer 360 in Silent Mode, 93
- After You Upgrade, 93

Upgrade Overview

A Customer 360 environment can include a development environment, a test environment, and a production environment. You must upgrade each of these environments. As a best practice, upgrade your development environment first. Identify and resolve any upgrade issues. After you successfully upgrade the development environment, you can upgrade the test and production environments with a high degree of confidence.

You can directly upgrade yourCustomer 360 version from the following previous Customer 360 versions installed:

- 10.4 including any hotfix
- 10.5 including any hotfix

You can upgrade the Customer 360 application in console or silent mode. When you upgrade Customer 360 in console mode, you need manual interaction to provide the required upgrade options during the process of upgrade. When you upgrade Customer 360 in silent mode, you do not need any interaction during the process of upgrade.

To learn about changes to the system requirements, review the Product Availability Matrix. You can find the Product Availability Matrix on Informatica Network:
https://network.informatica.com/community/informatica-network/product-availability-matrices/overview

Before You Upgrade

Before you upgrade Customer 360, upgrade the Multidomain MDM installation and back up the Customer 360 installation files.

1. Before you upgrade to *Multidomain MDM* , ensure that all the users submit the drafts records in Customer 360.
2. Upgrade the Multidomain MDM installation to a supported version.
3. Back up the Customer 360 Operational Reference Store.
4. Back up the Customer 360 library and configuration files.
5. Extract the latest Customer 360 application archive file.
6. Copy the Customer 360 configuration files to the `tcr` folder.
7. Configure the application properties file.
8. Configure the base URL for the business entity services.
9. Optionally, if you use the data model based on the Party Role table, migrate to the data model based on the Party table.
10. If you install Supplier 360 in the same environment, configure the log file path.
11. Before you upgrade Customer 360, ensure that you download and apply the Emergency Bug Fixes (EBF).
12. Validate the Customer 360 database schema and fix all the errors in the validation report.

Managing Drafts

Customer 360 deprecates its legacy draft creation capability and uses the draft creation capability from Multidomain MDM. Before you upgrade to *Multidomain MDM 10.4 HotFix 3*, ensure that the Customer 360 users submit their drafts. During upgrade, you can select to permanently delete all drafts.

Note: After you upgrade to *Multidomain MDM 10.4 HotFix 3*, you cannot retrieve the draft records.

Downloading and Applying Emergency Bug Fixes

Before you upgrade Customer 360, use the schema validation utility to validate the database schema for any errors. The validation utility generates a validation report that lists all the errors in the database and fixes all the errors in the validation report. To view all the errors in the validation report, ensure that you download and apply the following Emergency Bug Fixes (EBF):

- If you upgrade from Customer 360 10.3 HotFix 3, download and apply EBF-21675.
- If you upgrade from Customer 360 10.4, download and apply EBF-21761.
- If you upgrade from Customer 360 10.4 HotFix 1, download and apply EBF-21762.
- If you upgrade from Customer 360 10.4 HotFix 2, download and apply EBF-21763.
- If you upgrade from Customer 360 10.4 HotFix 3, ensure that you fix all the domain validation errors.

You can download the Emergency Bug Fixes (EBF) from the following portal:

`http://tsftp.informatica.com`

Validating the Database Schema

Use the schema validation utility to validate the database schema for any errors. The utility identifies the customized Customer 360 resources and the custom objects that do not conform to the naming guidelines. Manually fix the errors before you upgrade Customer 360.

Note: The page is applicable for versions up to 10.4 HotFix 2 of MDM Customer 360.

1. Open a command prompt, and go to the following directory:

   `<MDM installation directory>/app/tcr/bin`

2. Run the following command:
   - On Windows. `validate-schema-tcr.bat`
   - On UNIX. `validate-schema-tcr.sh`
3. At the prompts, enter the following parameters:

Parameter	Description
MDM Hub installation directory	Fully qualified path to the directory where you installed MDM Hub.
MDM - Customer 360 installation directory	Fully qualified path to the directory that contains the Customer 360 files.
Customer 360 Operational Reference Store	Database ID of the Customer 360 Operational Reference Store.
MDM Hub user name	MDM Hub user name to access the Customer 360 Operational Reference Store.
Password	Password for the user name.
WebLogic Console Password	Required if you use the WebLogic application server.

4. To validate another database schema, press **y**, and enter the following parameters:

Parameter	Description
Customer 360 Operational Reference Store	Database ID of the Customer 360 Operational Reference Store.
MDM Hub user name	MDM Hub user name to access the Customer 360 Operational Reference Store.
Password	Password for the user name.

5. If you do not want to validate another database schema, press **n**.

Upgrading Multidomain MDM

Upgrade the Multidomain MDM environment to a supported version.

For more information about upgrading Multidomain MDM, see the *Multidomain MDM Upgrade Guide*.

Note: When you upgrade Multidomain MDM, ensure that the date and time match in the database server and the Hub Server.

Backing Up the Customer 360 Operational Reference Store and Database

Before you upgrade Customer 360, ensure that you perform the following tasks:

- Back up the Customer 360 Operational Reference Store.
- Export the Customer 360 database to a change list XML file. After you upgrade Customer 360, use the change list XML file to restore the extended data model objects. Use the Repository Manager tool to export the database.

Backing Up the Customer 360 Library and Configuration Files

Before you back up the Customer 360 library and configuration files, ensure that you close the Customer 360 user interface and the Provisioning tool.

1. Go to the following directory:

   `<MDM Installation Directory>/app`

2. Rename the `tcr` folder.

   For example, rename the folder to `tcr_backup`.

Extracting the Customer 360 Application Archive File

You receive the Customer 360 application as an archive file. Create the following directory structure and extract the contents of the Customer 360 archive file into it:

`<MDM Installation Directory>/app/tcr`

The extracted content contains the following files and folders:

File or Folder Name	Description
`batchgroup`	Contains the JAR file for the silent installation process.
`bin/`	Contains installation, upgrade, and database schema validation utilities.
`bpm/`	Contains the ActiveVOS default business process in a deployable format. The folder contains the following sub-folders: - `pre_c360_10.2_hf2/`. Directory that contains the default business processes for the data model based on the Party Role table. - `source/`. Contains the source files for business processes. - `InfaCustomer360Workflow.bpr`. Contains the default business process for the data model based on the Party table.
`config/`	Contains configuration properties files.
`CustomerB2BPortal`	Contains the files of the preconfigured Customer Portal for Organization business entity.
`data/`	Contains the scripts for inserting lookup or reference data.
`datamodelmigration/`	Contains the scripts and the change lists required to migrate from the data model based on the Party Role table to the data model based on the Party table.

File or Folder Name	Description
docs/	Contains the data dictionary. The folder contains the following files and sub-folders: - C360_104HF1_DataDictionary_en.pdf. Data dictionary for the data model based on the Party table. - pre_c360_10.2_hf2. Directory that contains the data dictionary for the data model based on the Party Role table.
email-config	Contains the subdirectories that contain configuration files for customer portal email configuration. Following are the list of the subdirectories: - templates/. Contains the avos-templates with email body text templates for ActiveVOS. - emailConfig.xml. File containing the configuration properties for email templates.
extensions	Contains the files for Customer 360 extensions.
externalcall	Contains the folders and files for SOAP services.
hub/	Contains the subdirectories that contain the database schema and the configuration files to deploy. Following are the list of the subdirectories: - change-xml/. Contains the MDM Hub metadata including components, such as landing tables, lookup tables, staging tables, base objects, and match and merge rules, cleanse functions, component instances, business entities, and business entity services. - cocs-config-xml/. Contains configuration files for the business entities and business entity services. - delta_change_xml/. Contains the newly added MDM Hub metadata. - entity360config-xml/. Contains copies of the Entity 360 component instance definitions that ship with Multidomain MDM.
idd/	Contains the message and error bundle files.
lib/	Directory for the external libraries. Copy the JDBC driver files for your database to the lib directory.
localizationScript	Contains the scripts for localizing labels and error messages.
lookuplocalization/	Contains the localized lookup data.
notices/	Contains the legal copyright file.
reporting/	Contains datamart folder which contains the data mart service and the chart configurations. The pre_c360_10.2_hf2/ directory in the datamart folder contains the data mart service for the data model based on the Party Role table.
resources/	Contains the resource bundle.properties files for each of the supported locales.
upgrade/	Contains the supporting files for the upgrade utility.
user_exit/	Contains the MDM Hub user exit archive.

File or Folder Name	Description
was	Contains the Provisioning tool user interface for a WebSphere environment.
bundleLocalization.jar	JAR file for localization.
Customer360ExternalCall.ear	File for SOAP service to validate the customer data.
domain-validation.jar	JAR file for validating Customer 360 domain.
email-config-util	JAR file for email configuration.
MDMAppsServices.war	File that contains Customer 360 services.
mdmappsview-ear.ear and uiwebapp-ear.ear	Files that contain Customer 360 user interface JBoss.
productversion.jar	JAR file for the product version.
provisioning-ear.ear	File that contains the Provisioning tool user interface.
store-credentials.jar	File that contains credential details.

Copying Compatible JAR File

To successfully upgrade and ensure compatibility with latest JDK, complete the following steps:

1. Navigate to the following directory:

   `<MDM installation directory>/app/tcr/`
2. Copy the `domain-validation.jar` file and copy to the following directory:

   `<MDM installation directory>/hub/server/lib/`
3. Restart the application server.
4. Validate the Supplier 360 Operational Reference Store (ORS) from the Hub Console.

   If the domain validation passes successfully, proceed with the upgrade of Supplier 360 components.

Copying the Customer 360 Configuration Files

To retrieve the existing Customer 360 configuration, copy the backed up Customer 360 configuration files to the newly created `tcr` folder.

1. Copy all the files in the following directory:

   `<MDM Installation Directory>/app/<renamed tcr folder>/config`
2. Paste the files in the following directory:

   `<MDM Installation Directory>/app/tcr/config`

   Note: You must replace the existing files.

Configuring the Application Properties File

In the `application.properties` file, configure the properties related to file import and ActiveVOS.

1. Navigate to the following directory:

   `/app/tcr/config`

2. Open the `application.properties` file in an editor, and configure the following properties:

   activevos.datasource.url

   Specifies the JNDI name of the ActiveVOS data source.

   bulkimport.fileupload.cleanupMinutes

   Optional. Manually add the `bulkimport.fileupload.cleanupMinutes` property. Specifies the number of minutes to persist the import file in the specified directory after the import process is completed. The import process deletes the file after the specified time period. Default is 1 minute.

3. Save the file.

Configuring the bes-client Properties File

Specify the connection properties in the `bes-client.properties` file.

1. Navigate to the following directory:

   `<MDM installation directory>/app/tcr/config`

2. Open the `bes-client.properties` file in an editor.
3. Specify the following connection properties:

Property	Description
siperian-client.protocol	Communication protocol that you want to use. Default is HTTP. Do not change the default value.
bes-client.http.url	URL to connect to the Hub Console. For example, `http://<Host>:<Port number>/cmx`

4. Save the file.

Configuring the Properties File for Silent Installation

If you want to install the Customer 360 application without user interaction in silent mode, configure the `C360_silent_installer.properties` file. When you perform the silent installation, the installer reads the `C360_silent_installer.properties` file to determine the installation options. Ensure that you provide correct settings in the properties file.

1. Go to the following directory:

   `<MDM installation directory>/app/tcr/config`

2. Open the `C360_silent_installer.properties` file.
3. Set the values for the required parameters in the `C360_silent_installer.properties` file in a text editor.

4. To validate and upgrade multiple Customer 360 Operational Reference Stores, add the list of Operational Reference Stores in the `ORS_ID` property.

   For example, `ORS_ID=Schema1,Schema2`

5. To access the Customer 360 Operational Reference Stores, add the following properties for each Operational Reference Store:

Property	Description
`<ORS ID 1>.USERNAME`	User name to access the Operational Reference Store 1.
`<ORS ID 1>.PASSWORD`	Password for the user name.

6. Comment all the properties added for upgrade.
7. Save the file.

Configuring the Base URL for the Business Entity Services

If the base URL is not previously configured, configure the base URL for the business entity services to access the Hub Server.

1. Open the following file in a text editor:

   `<MDM Installation Directory>/app/tcr/config/mdmapps-config.properties`

2. Configure an appropriate URL for the `base.url` property to access the Hub Server.

   Use one of the following formats for the base URL based on your environment:

   - `http://<Host>:<Port>`
     `Host` indicates the host name or the IP address of the machine on which you deploy Customer 360 and `Port` indicates the port number on which the host listens.
   - `http://<Domain name>`
     `Domain name` indicates the fully qualified domain name of the machine that hosts Customer 360.

Migrating to the Data Model Based on the Party Table

Customer 360 versions 10.2 HotFix 2 and later use the data model based on the Party table. If you upgraded to 10.2 HotFix 2 or use Customer 360 version 10.2 HotFix 1 or 10.2, then Customer 360 uses the data model based on the Party Role table. You can now migrate from the data model based on the Party Role table to the data model based on the Party table.

Note: If you do not want to migrate to the data model based on the Party table, you can continue to use the data model based on the Party Role table.

1. Promote the `update_foreign_keys.change.xml` change list.
2. Validate the Customer 360 Operational Reference Store.
3. Update foreign keys and recalculate the best version of truth for the records.
4. Validate the Customer 360 Operational Reference Store.
5. Promote the `party_updates.change.xml` change list.
6. Validate the Customer 360 Operational Reference Store.
7. If you extended the data model, import the data model extensions.
8. Reindex the data.

9. Deploy the updated Customer 360 components.
10. Reconfigure the values for the Full Name row in the Hierarchy Manager.
11. Remove the post-merge user exit.
12. Delete the shards and the ZooKeeper data directory.

Step 1. Promote the update_foreign_keys.change.xml Change List

The `update_foreign_keys.change.xml` change list contains information about the foreign keys for the data model based on the Party table. Use the Repository Manager tool to promote the change list.

1. In the **Configuration** workbench of the Hub Console, click **Repository Manager**.
2. On the **Promote** tab, click **Change List**.
3. Select the Customer 360 Operational Reference Store as the target repository.
4. Click **Open**, and browse to the following directory:

   `<MDM installation directory>/app/tcr/datamodelmigration/hub/change-xml`
5. Select the `update_foreign_keys.change.xml` file, and click **Open**.

   The Repository Manager tool loads the list of changes that the change list contains.
6. Click **Apply Changes**.

   The Repository Manager tool prompts you to select a rollback strategy in the event of a failure.
7. Select the rollback strategy that you want to use, and click **OK**.

   A confirmatory message appears.
8. Click **Yes**.

Step 2. Validate the Customer 360 Operational Reference Store

After you apply the `update_foreign_keys.change.xml` change list, validate the Customer 360 Operational Reference Store.

1. In the **Configuration** workbench of the Hub Console, click **Repository Manager**.
2. On the **Validate** tab, select the Customer 360 Operational Reference Store.
3. Click **Validate**.

   The **Select Validation Checks** dialog box appears.
4. Select the validation checks to perform, and click **OK**.

   The Repository Manager tool validates the repository and displays any issues in the **Issues Found** pane.
5. To fix a repairable issue, perform the following tasks:
   a. Select the issue that you want to repair, and click **Repair**.

      A warning message appears.
   b. Click **OK**.

Step 3. Update Foreign Keys and Recalculate Best Version of Truth

After you apply the `update_foreign_keys.change.xml` change list, update the foreign keys and recalculate the best version of truth for the records. After you recalculate the best version of truth, verify whether the best version of truth is successfully calculated for all the records.

1. Open a command prompt, and go to the following directory:

   `<MDM installation directory>/app/tcr/datamodelmigration/bin`
2. To update the foreign keys, run the following command:
   - On Windows. `migrate_to_party_model.bat update_foreign_keys`
   - On UNIX. `migrate_to_party_model.sh update_foreign_keys`
3. To recalculate the best version of truth for the records, run the following command:
   - On Windows. `migrate_to_party_model.bat update_BVT`
   - On UNIX. `migrate_to_party_model.sh update_BVT`
4. To verify whether the best version of truth is calculated for all the records, perform the following steps:
   a. Run the following command:
      - On Windows. `migrate_to_party_model.bat verify_data_update`
      - On UNIX. `migrate_to_party_model.sh verify_data_update`

   b. Open the following log file:

      `<MDM installation directory>/app/tcr/datamodelmigration/log/verify_data_update.log`
      The log file displays the number of records that failed the best version of truth calculation for each base object. The value 0 indicates that the best version of truth is calculated for all the records.

      The following sample log entry indicates that the best version of truth is calculated for all the records in the C_BO_PRTY_RLE_STCK_EXCH table:

      ```
      0 rows affected
      TABLE_NAME,COUNT(1)
      C_BO_PRTY_RLE_STCK_EXCH,0
      ```
5. To delete the configuration details that are irrelevant to the data model based on the Party table, run the following command:
   - On Windows. `migrate_to_party_model.bat delete_configuration`
   - On UNIX. `migrate_to_party_model.sh delete_configuration`

Step 4. Validate the Customer 360 Operational Reference Store

After you recalculate the best version of truth, validate the Customer 360 Operational Reference Store.

1. In the **Configuration** workbench of the Hub Console, click **Repository Manager**.
2. On the **Validate** tab, select the Customer 360 Operational Reference Store.
3. Click **Validate**.

   The **Select Validation Checks** dialog box appears.
4. Select the validation checks to perform, and click **OK**.

   The Repository Manager tool validates the repository and displays any issues in the **Issues Found** pane.
5. To fix a repairable issue, perform the following tasks:
   a. Select the issue that you want to repair, and click **Repair**.

      A warning message appears.

b. Click **OK**.

Step 5. Promote the party_updates.change.xml Change List

The `party_updates.change.xml` change list contains changes related to the data model based on the Party table. Use the Repository Manager tool to promote the change list.

1. In the **Configuration** workbench of the Hub Console, click **Repository Manager**.
2. On the **Promote** tab, click **Change List**.
3. Select the Customer 360 Operational Reference Store as the target repository.
4. Click **Open**, and browse to the following directory:

   `<MDM installation directory>/app/tcr/datamodelmigration/hub/change-xml`
5. Select the `party_updates.change.xml` file, and click **Open**.

   The Repository Manager tool loads the list of changes that the change list contains.
6. Click **Apply Changes**.

   The Repository Manager tool prompts you to select a rollback strategy in the event of a failure.
7. Select the rollback strategy that you want to use, and click **OK**.

   A confirmatory message appears.
8. Click **Yes**.

Step 6. Validate the Customer 360 Operational Reference Store

After you apply the `party_updates.change.xml` change list, validate the Customer 360 Operational Reference Store.

1. In the **Configuration** workbench of the Hub Console, click **Repository Manager**.
2. On the **Validate** tab, select the Customer 360 Operational Reference Store.
3. Click **Validate**.

   The **Select Validation Checks** dialog box appears.
4. Select the validation checks to perform, and click **OK**.

   The Repository Manager tool validates the repository and displays any issues in the **Issues Found** pane.
5. To fix a repairable issue, perform the following tasks:
   a. Select the issue that you want to repair, and click **Repair**.

      A warning message appears.
   b. Click **OK**.

Step 7. Import the Data Model Extensions

If you extended the data model, manually import the extended objects. To identify the extended objects, use the change list that you generated after exporting the Customer 360 database. Use the Repository Manager tool to import the change list.

Step 8. Reindex the Data

To reindex the data, run the Initially Index Smart Search Data batch job on all the base objects that contribute to each business entity.

Use the Hub Console to run the Initially Index Smart Search Data batch job. For more information about running the Initially Index Smart Search Data batch job, see the *Multidomain MDM Configuration Guide*.

Step 9. Deploy the Updated Customer 360 Components

After you reindex the data, deploy the updated Customer 360 components.

1. Open a command prompt, and go to the following directory:

   `<MDM installation directory>/app/tcr/datamodelmigration/bin`
2. Run the following command:
   - On Windows. `migrate_to_party_model.bat deploy_c360_components`
   - On UNIX. `migrate_to_party_model.sh deploy_c360_components`

   The command deploys the updated Customer 360 components.
3. To verify whether all the components are successfully deployed, perform the following tasks:
   a. Use a text editor to open the following file:

      `<MDM installation directory>/hub/server/conf/c360-versioninfo.xml`

   b. Check the value of the `variant` parameter.

      The value `party` indicates that all the components are successfully deployed. If you do not find the `variant` parameter, then one or more components are not properly deployed. Run the command again to redeploy all the components.

Step 10. Reconfigure the Values for the Full Name Column in the Hierarchy Manager

After you deploy the Customer 360 components, reconfigure the label and tooltip values for the Full Name column of the Person and Organization business entities.

1. In the Model workbench of the Hub Console, click **Hierarchies**.
2. Acquire a write lock.
3. In the navigation pane of the Hierarchies tool, select **Profiles** > **Default** > **Entity Objects** > **Party** > **Organization**.
4. In the **HM Packages** section, reset the **Label** and **Tooltip** values to 0 for the **Full Name** column.
5. In the navigation pane of the Hierarchies tool, select **Profiles** > **Default** > **Entity Objects** > **Party** > **Person**.
6. In the **HM Packages** section, reset the **Label** and **Tooltip** values to 0 for the **Full Name** column.
7. Click **Save**.

Step 11. Remove the Post-Merge User Exit

The data model based on the Party table does not use the post-merge user exit, so you must remove it.

1. In the **Utilities** workbench of the Hub Console, click **User Object Registry**.
2. Acquire a write lock.
3. In the navigation pane, select **User Exits**.

   The **User Exits** page appears.
4. Based on the listed user exit, perform one of the following tasks:
   - If the listed user exit is **Post Merge User Exit**, click **Remove**, and then click **OK**.
   - If you see any other user exit, perform the following tasks:
     1. Browse to the JAR file that contains the user exits.
     2. Extract the JAR file, and remove the `PostMergeImplementation.class` file from the JAR file.
     3. Export the remaining user exit classes to a JAR file.
     4. In the **User Exits** page, click **Add**.

        The **Add User Exit** dialog box appears.
     5. Click **Browse**, browse to the updated JAR file that contains the user exits, and click **Open**.
     6. Optionally, enter a description.
     7. Click **OK**.

Step 12. Delete the Shards and the ZooKeeper Data Directory

After you remove the post-merge user exit, delete the shards and the ZooKeeper data directory. When you configure the Process Server for search, you specify the path to the ZooKeeper data directory.

You can find the shards in the following directory: `<MDM installation directory>/hub/cleanse/solr`

Configuring the Log File Path

Specify the path of the log file in the `mdmapps-log4j.properties` file.

1. Navigate to the following directory:

   `<MDM installation directory>/app/tcr/config`
2. Open the `mdmapps-log4j.properties` file in an editor.

3. Specify the following log file properties:

Property	Description
`appender.file.fileName`	Path to the log file. For example, `appender.file.fileName=<MDM installation directory>/mdmapplogs/mdmapps.log` **Note:** If you plan to install Supplier 360 in the same environment, ensure that you specify a location that is external to both the applications. A common file stores the logs for both the applications.
`appender.rolling.filePattern`	Pattern for the log file name. For example, `appender.rolling.filePattern=<MDM installation directory>/mdmapplogs/mdmapps-%i.log` You can refer to the following format of the log file name: `<MDM installation directory>/mdmapplogs/mdmapps-1.log`

4. Save the file.

Upgrading Customer 360 in Console Mode

Before you upgrade Customer 360, ensure that you validate the Customer 360 Operational Reference Store through the Repository Manager tool of the Hub Console.

1. Open a command prompt, and go to the following directory:

   `<MDM installation directory>/app/tcr/bin`

2. Run the following command:
   - On Windows. `upgrade-tcr.bat`
   - On UNIX. `upgrade-tcr.sh`
3. At the prompts, enter the following parameters:

Parameter	Description
MDM Hub installation directory	Fully qualified path to the directory where you installed MDM Hub.
MDM - Customer 360 installation directory	Fully qualified path to the directory that contains the Customer 360 files.
Application Server	Name of the application server in lowercase. For example, weblogic, jboss, websphere.
avos console username	User name with administrative privileges to access the ActiveVOS Console.

Parameter	Description
avos console password	Password of the ActiveVOS Console user name.
PIM portal username	Product 360 user name that you entered in the previous installation. If you do not want to enable Product 360, press **Enter**.
PIM portal password	Product 360 password that you entered in the previous installation.
Customer 360 Operational Reference Store	Database ID of the Customer 360 Operational Reference Store.
MDM Hub user name	MDM Hub user name to access the Customer 360 Operational Reference Store.
Password	Password for the user name.

Note: If you use a WebLogic application server, you are prompted to enter the WebLogic console password.

4. To upgrade another validated database schema, perform the following steps:
   a. Press **y**.
   b. Enter the following parameters:

Parameter	Description
Customer 360 Operational Reference Store	Database ID of the Customer 360 Operational Reference Store.
MDM Hub user name	MDM Hub user name to access the Customer 360 Operational Reference Store.
Password	Password for the user name.

   The database schema upgrade starts.

   c. To upgrade another validated database schema, perform steps a through b.
5. If you do not want to upgrade another database schema, press **n**.
6. To start the Customer 360 upgrade, press **y**.
7. After you successfully upgrade Customer 360, restart the application server.

After you upgrade, if you find any errors on the **Domain Results** tab of the Repository Manager tool, you must fix the errors. After fixing all the validation errors, run the upgrade script to upgrade Customer 360 and the database schemas.

For more information, see the *Multidomain MDM Repository Manager Guide*.

Upgrading Customer 360 in Silent Mode

You can upgrade Customer 360 in silent mode without any user interaction. Before you upgrade Customer 360 in silent mode, ensure that you configure the `C360_silent_installer.properties` file.

1. Open a command prompt, and navigate to the following directory:

   `<MDM installation directory>/app/tcr/bin`

2. Run the following command:

   - On Windows. `upgrade-tcr.bat silent <MDM installation directory>\app\tcr\config \C360_silent_installer.properties`
   - On UNIX. `./upgrade-tcr.sh silent <MDM installation directory>/app/tcr/config/ C360_silent_installer.properties`

   Note: The upgrade script runs in the background. The process can take a while to complete. After the installation is complete, review the messages to ensure the successful upgrade of Customer 360.

After You Upgrade

After you upgrade Customer 360, perform the following tasks:

- Install the Customer 360 consent management extension.
- Verify the version information file.
- Run the lookup scripts to update the reference data.
- Import the localized lookup data into the Customer 360 database.
- Perform the steps to improve the performance of the bulk data import.
- Update all the charts or specific chart in the Customer 360 database.
- Add the customizations in the custom dashboard.
- Validate the Customer 360 database and fix all the domain errors.
- Log in to Customer 360 with administrator credentials and test Customer 360.

Updating the Reference Data

After you upgrade Customer 360, update the reference data in the tables.

Before you update the reference data, if you have customized the reference data, update the lookup script to include the customization.

1. Open a command prompt, and navigate to the following directory:

   `<MDM installation directory>/app/tcr/data/reference-data`

2. Run one of the following lookup scripts based on the database you use:

   - For Oracle. `C360_lookup_script_oracle.sql`
   - For Microsoft SQL Server. `C360_lookup_script_MSSQL.sql`
   - For IBM DB2. `C360_lookup_script_DB2.sql`

3. To verify whether the lookup records are loaded successfully, perform the following tasks:
   a. In the Utilities workbench of the Hub Console, click **Batch Group**.
   b. Expand **BG_All_Lookup_Load**, and select **Control & Logs**.
   c. In the Logs for each job table, review the **Status** column to verify that the load is successful.
   d. If the load is unsuccessful, try running the load. Select **BG_All_Lookup_Load** and click **Execute**.

Installing Customer 360 Consent Management Extension

After you install Customer 360, you can install the consent management extension.

1. Go to the following directory:

   `<MDM installation directory>/app/extension/GDPR/bin`
2. Run one of the following scripts:
   - On Windows. `install.bat`
   - On Linux. `install.sh`
3. At the prompts, enter the following parameters:

Parameter	Description
MDM Hub installation directory	Fully qualified path to the directory where you installed MDM Hub.
MDM - Customer 360 installation directory	Fully qualified path to the directory that contains the Customer 360 files.
Application Server	Name of the application server in lowercase.
avos console username	ActiveVOS Console user name.
avos console password	ActiveVOS Console password.

Note: If you use a Weblogic application server, you are prompted to enter the Weblogic console password.

Verifying the Version Information File

After you successfully upgrade Customer 360, verify the version information in the following file: `<infamdm installation directory>/hub/server/conf/c360-versioninfo.xml`.

If you use party role data model, in the `c360-versioninfo.xml` file, ensure that the value of the `variant` property is `partyRole`.

Inserting Reference Data for Customer 360 Consent Management Extension

If you plan to install the consent management extension, you can populate the tables with reference data related to consent management.

1. Open a command prompt, and navigate to the following directory:

   `<MDM Installation Directory>/app/tcr/extensions/GDPR/data/reference-data`

2. Run one of the following lookup scripts based on the database you use:
   - For Oracle. `C360_GDPR_lookup_script_Oracle.sql`
   - For Microsoft SQL Server. `C360_GDPR_lookup_script_MSSQL.sql`
   - For IBM Db2. `C360_GDPR_lookup_script_DB2.sql`
3. To verify whether the lookup records are loaded successfully, perform the following tasks:
   a. In the Utilities workbench of the Hub Console, click **Batch Group**.
   b. Expand **BG_All_Lookup_Load**, and select **Control & Logs**.
   c. In the Logs for each job table, review the **Status** column to verify that the load is successful. The **Total records** column shows the number of records added. The columns to the right of the **Total records** column displays zeros if all the records load successfully.
   d. If the load is unsuccessful, try running the load. Select **BG_All_Lookup_Load** and click **Execute**.

Updating Address Verification License Key

After you upgrade, ensure that you update the Address Verification license key from `ADV5` to `ADV6`.

Use the MDM Hub with Address Verification to validate and verify postal addresses, email addresses, and phone numbers. You can configure parameters for the type of address verification you want to perform.

For more information about updating `ADV6` and adding different parameters, see "Informatica Data as a Service" on page 20.

Importing the Localized Lookup Data

Customer 360 installation files include the localized lookup data. If you use a localized environment, import the localized lookup data into the database before you use Customer 360.

1. Copy the following files from the `<MDM installation directory>/hub/server/lib` directory to the `<MDM installation directory>/app/tcr/lookuplocalization/lib` directory:
   - `log4j-core-2.17.2.jar`
   - `log4j-1.2-api-2.17.2.jar`
   - `log4j-api-2.17.2.jar`
   - For Oracle. `ojdbc8.jar`
   - For IBM DB2. `db2jcc.jar`
   - For Microsoft SQL Server. `sqljdbc4.jar`
2. Go to the following directory:

   `<MDM installation directory>/app/tcr/lookuplocalization/bin`

3. Run the following command:

- On UNIX. `./lookup_localization.sh`
- On Windows. `lookup_localization.bat`

4. At the prompts, enter the following parameters:

Parameter	Description
Database type	Type of database that you use. Use one of the following values: - Oracle - DB2 - MSSQL
User name	User name to access the Operational Reference Store database.
Password	Password for the user name.
Operational Reference Store database host name	Name of the host that runs the Operational Reference Store database.
Operational Reference Store database port number	Port number that the database listener uses.
Database name	For IBM DB2 and Oracle only. Name of the IBM DB2 database or Oracle service.

   The localized lookup data is imported into the staging tables.
5. In the Utilities workbench of the Hub Console, click **Batch Group**.
6. Expand **Localized Lookup Data Load**, and select **Control & Logs**.
7. Click **Execute**.

   The localized lookup data is imported into the base objects.

Improving Performance of Bulk Data Import

When you import a file in a workflow-enabled environment, the workflow triggers the approval process in parallel. The parallel processing of workflow might affect the performance of the import process when you import more than 5000 records with child records. To improve the performance of the import process in a workflow-enabled environment, you can now set a wait time for the workflow process to trigger. Based on the wait time, the workflow process gets triggered and avoids parallel processing.

1. Go to the following directory:

   `<MDM installation directory>/app/tcr/config/`
2. In a text editor, open the `c360-portal-workflow-config.xml` file.

3. Update the following properties and save the file:

Property	Description
`<property name="portal.fileImport.wait">P0Y0M0DT2H0M0S</property>`	Optional. Wait time for the workflow to trigger after the import process starts. For example, if the wait time is 2 hours 30 minutes, then the workflow triggers after 2 hours 30 minutes after the import process starts. You can specify the wait time in the following format: `P<n>Y<n>M<n>DT<n>H<n>M<n>S` - P - Indicates portal. - Y - Indicates number of years. - M - Indicates number of months. - D - Indicates number of days. - T - Indicates time. - H - Number of hours. - M - Number of minutes. - S - Number of seconds. For example, the value `P0Y0M0DT2H30M0S` indicates 2 hours and 30 minutes. Ensure that you set the `portal.fileImport` property to `Y` for the wait time to be considered. Default is 2 hours. **Note:** Informatica recommends to use 1 hour 30 minutes for 5000 records and 2 hours 30 minutes for 10000 records.
`<property name"portal.fileImport">Y</property>`	Indicates whether to enable wait time for the workflow to trigger. Set to `Y` to enable the wait time. The wait time improves the performance of import process when you import more than 5000 records. Use this option during the file import or initial data load process. After the data load is complete, change the value from `Y` to `N`. Default is `N`.

4. After you make changes to the properties, run one of the following redeploy scripts to view the changes:
   - On UNIX. `redeploy.sh`
   - On Windows. `redeploy.bat`

   Note: Alternatively, you can log in to the ActiveVOS console, navigate to the **Catalog** > **Resources** > **Other** section to update the properties in the `c360-portal-workflow-config.xml` file.
5. Go to the following directory:

   `<MDM installation directory>/hub/server/resources`
6. In a text editor, open the `cmxserver.properties` file.

7. Update the following properties and save the file:

Property	Description
`cmx.server.be-import.task-limit`	Specifies the maximum number of records users can import for the task approval workflow to trigger. Set `cmx.server.be-import.task-limit=10000` for users to import up to 10000 records and the task approval workflow to trigger. If a user attempts to import more than 10000 records, the task approval workflow does not trigger and displays an error.
`cmx.server.find-replace.task-limit`	Specifies the maximum number of replaced records that trigger the task approval workflow. Set `cmx.server.find-replace.task-limit=1000` for users to replace up to 1000 records and the task approval workflow to trigger. If a user attempts to import more than 1000 records, the task approval workflow does not trigger and displays an error.

8. Restart the application server.

Updating the Charts

After you upgrade Customer 360, you can update a specific chart or all the charts.

1. Open a command prompt, and go to the following directory:
   - For the data model based on the Party table. `<MDM installation directory>/app/tcr/reporting/datamart`
   - For the data model based on the Party Role table. `<MDM installation directory>/app/tcr/reporting/datamart/pre_c360_10.2_hf2`
2. To update a specific chart, run the following command:

   ```
   java -jar populate_datamart.jar exec <Report name>
   ```
3. To update all the charts, run the following command:

   ```
   java -jar populate_datamart.jar
   ```

Updating the Custom Dashboard

If you use custom dashboard, manually add the customizations to the dashboard.

1. In the Provisioning tool, click **Configuration** > **Application Editor**.

   The **Application Editor** appears.
2. In the **Applications** list, select **MDMApps Application Configuration**, and then select **MDMAppsWidgetConfiguration**.
3. In the **Properties** panel, add the XML code for the customizations.

   Note: You can refer to the properties of **C360WidgetConfiguration** to identify the customizations.
4. Click **Apply**.
5. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A confirmation dialog box appears that prompts you to publish or review the changes.

b. Review the changes or publish without a review.

- To publish without a review, click **Publish**.
- To publish after a review, click **Review Changes** and follow the instructions that appear on the screen.

Validating the Customer 360 Database Schema

After you upgrade, you must validate the Customer 360 schema and fix all the errors. To find and resolve the errors, use the Repository Manager tool in the Hub Console.

For more information about validating the database schema, see the *Informatica Multidomain MDM Repository Manager Guide*.

Configuring the Content Security Policy to View the Dashboard

Configure the content security policy in the Provisioning tool to view data on the dashboard that you see on the Home page.

1. Log in to the Provisioning tool, and select the Customer 360 Operational Reference Store.
2. Click **Configuration** > **General Settings**, and then select **Content Security Policy**.
3. Click **Content Security Policy**.

   The tree view panel and the content security policy panel display the content security policy options.
4. In the **Content Security Policy** panel, click **Add Content Security Policy Tag**.
5. From the **Policy Tag** list, select **script-src-elem**.
6. Enter the following required values for the policy tag:

   - `'self'`
   - `'unsafe-inline'`
   - `*`

   Use the **Add** button to add multiple values for a policy tag. Use the **-** button to delete values.
7. To apply the updated content security policy to your system, click **Apply**.
8. Click **Publish**.
9. Review the changes and click **Confirm**.

Test Customer 360

You can log in to Customer 360 with your administrative user credentials.

1. Use the following URL format to access Customer 360:

   ```
   http://<MDM Server host name>:<MDM Server port number>/mdmapps/
   com.informatica.tools.mdm.web.auth/login
   ```
2. Log in with your user credentials.
3. If prompted to select an application, select **Customer 360**.

   The **Home** page appears.

CHAPTER 8

Troubleshooting

This chapter includes the following topics:

- Troubleshooting the Customer 360 Installation, 100
- Troubleshooting the Customer 360 Upgrade, 100
- Troubleshooting the Customer 360 Configuration, 103

Troubleshooting the Customer 360 Installation

If you encounter any issues when you install Customer 360, use the following information to troubleshoot the issues.

The installer does not update the JNDI path for JBoss.

During reinstallation of Customer 360, if you change the installation directory, the JNDI path in the JBoss application server does not reflect the change.

Restart the JBoss application server to get the changed installation directory path.

Troubleshooting the Customer 360 Upgrade

If you encounter any issues when you upgrade Customer 360, use the following information to troubleshoot the issues.

The Customer 360 upgrade fails.

When you run the upgrade script, the upgrade fails with the following error:

```
Error Message ---> Object 'RESOURCE.HM_RELATIONSHIP_TYPE.ContactEmployee' already
exists. Object cannot be added.
```

To resolve this issue, perform the following tasks:

1. Go to the following directory:

   ```
   <MDM installation directory>/app/tcr/hub/delta_change_xml/10.2 HF3 EBF1/pre_c360_10.2_hf2
   ```

2. Delete the `TCR-2737.change.xml` file.
3. Run the upgrade script.

The Customer 360 upgrade fails without restoring the database to the pre-upgrade state.

To resolve this issue, perform the following tasks:

1. Go to the following directory:
   `<MDM installation directory>/hub/`
2. Open the `upgrade.log` file and identify the errors.
3. Fix the issues identified in the `upgrade.log` file.
4. Go to the following directory:
   `<MDM installation directory>/app/tcr/hub/delta_change_xml/10.4 HF2`
5. Delete the `<name of the changelist>.change.xml` files that were successfully applied.
   The `upgrade.log` file contains the list of successfully applied change lists.
6. Run the upgrade script.

The Customer 360 upgrade fails while promoting the change list.

When you run the upgrade script, the upgrade fails with the following error:

```
Error Message ---> Object 'WRITE_CO_CHILD.CustomerOrg|CustomerOrgPortalContacts' already
exists. Object cannot be added. (addWriteCOChild[id:WRITE_CO_CHILD.CustomerOrg|
CustomerOrgPortalContacts,parentId:WRITE_CO_ROOT.CustomerOrg])
```

To resolve this issue, perform the following tasks:

1. Go to the following directory:
   `<MDM installation directory>/hub/`
2. Open the `upgrade.log` file and identify the errors.
3. Fix the issues identified in the `upgrade.log` file.
4. Go to the following directory:
   `<MDM installation directory>/app/tcr/hub/delta_change_xml/<upgrade version directory>`
5. Delete the `<name of the changelist>.change.xml` file because of which the upgrade failed.
   The `upgrade.log` file contains the `<name of the changelist>.change.xml` file because of which the upgrade failed.
6. Save the file.
7. Run the upgrade script.

The Customer 360 upgrade fails in a WebSphere environment.

In a WebSphere environment, when you run the upgrade script, the upgrade fails with the following error:

```
Error Message ---> unable to undeploy the provisioning.ear application. Application does
not exist.
```

The upgrade fails because of the change in the name of the `provisioning.ear` file.

To resolve this issue, perform the following tasks:

1. Log in to the WebSphere Application Server administrative console.
2. Click **Deployments**.
3. Undeploy the existing `provisioning-ear.ear` file.
4. Go to the following directory:
   `<MDM installation directory>/hub/server/`
5. Run the `patchInstallSetup` script.

On UNIX

```
patchInstallSetup.sh -Ddatabase.password=<MDM Hub Master database password> -
Davos.username=<ActiveVOS Console username> -Davos.password=<ActiveVOS Console
password> -Davos.jdbc.database.username=<ActiveVOS database username> -
Davos.jdbc.database.password=<ActiveVOS database password> -Dconsole.hostname=<host
name> -Dconsole.webport=<port>
```

On Windows

```
patchInstallSetup.bat -Ddatabase.password=<MDM Hub Master database password> -
Davos.username=<ActiveVOS Console username> -Davos.password=<ActiveVOS Console
password> -Davos.jdbc.database.username=<ActiveVOS database username> -
Davos.jdbc.database.password=<ActiveVOS database password> -Dconsole.hostname=<host
name> -Dconsole.webport=<port>
```

Note: For more information about running the `patchInstallSetup` script, see the *Multidomain MDM 10.4 HotFix 3 Installation Guide*.

After you upgrade to Customer 360 10.4 HotFix 3, the title of the person entity draft is saved with the business entity name and does not reflect the format you set.

To resolve this issue, perform the following tasks:

1. In the Provisioning Tool, click **Configuration** > **General Settings**, and then select **Metadata Localization**.
2. Click **Metadata Localization**.
3. In the Metadata Localization panel, under **Metadata Bundle**, click **Export**.
4. Extract the `besMetadata.zip` file to your local drive.
5. Open the `besMetadata.properties` file in a text editor.
6. Update the value of the following property from `{fullNm}` to `{fullNm} {lstNm}`:
   `CustomerPerson.label.existsFormat`
7. Similarly, update other localized files.
8. Compress all the localized properties file into a .zip file.
9. In the Metadata Localization panel, under **Metadata Bundle**, click **Import** and select the .zip file that you created.
   The metadata bundle is imported with the updated configuration.

After you upgrade to Customer 360 10.4 HotFix 3, if you click the title of a draft record, the "The configuration of business entity has changed. Some fields might not appear in the draft" message appears even though no configuration changes were made.

To resolve this issue, perform the following tasks:

1. In the Provisioning Tool, click **Configuration** > **Layout Designer**.
   The **Layout Designer** appears.
2. In the **Layout Categories** panel, select the layout for which the message appears in the draft workspace.
3. Click **Edit**.
   The **Design** page appears.
4. Click **Configure** icon of the data component in the workspace.
   The **Record Details Component Designer** dialog box appears.
5. In the **Attributes layout** section, find the **draftInd** attribute, and click the **Delete** icon.
6. Click **Done**.
   The **Design** page appears.
7. Click **Save**.

After you upgrade to Customer 360 10.4 HotFix 3, the Repository Manager tool does not display the Domain Results tab in the Hub Console, and the C_REPOS_CO_CS_CONFIG repository table has no entry for domain validation.

To resolve this issue, perform the following tasks:

1. Open a command prompt, and go to the following directory:
   `<MDM installation directory>/hub/server/conf`
2. Open the `C360_version.info` file and note down the following information:
   - start-install-time
   - end-install-time
   - internal-version
   - name
   - external-version
   - variant
3. Run the following REST API request with the information that you noted down from the `C360_version.info` file:
   `POST http://<hostname>:<port>/cmx/<database_id>/validation/domain/version/ProductVersion`

   The following request body format includes information that you noted from the `C360_version.info` file:

   ```
   <?xml version="1.0" encoding="UTF-8" standalone="yes"?>
   <ComponentVersion>
   <!--Use the information retrieved from the C360_version.info file-->
       <start-install-time>install_start_time</start-install-time>
        <end-install-time>install_end_time</end-install-time>
       <internal-version>internal_version</internal-version>
       <name>name_of_the_product</name>
       <external-version>external_version</external-version>
       <install-status>install_status</install-status>
       <variant>variant</variant>
   </ComponentVersion>
   ```
4. Click **Send**.
   After you send the REST API request, the Repository Manager tool displays the **Domain Results** tab in the Hub Console, and an entry for domain validation gets added to the `C_REPOS_CO_CS_CONFIG` repository table.

Troubleshooting the Customer 360 Configuration

If you encounter any issues when you configure Customer 360, use the following information to troubleshoot the issues.

I accidentally generated business entity schema for Customer 360.

In the Informatica Data Director Configuration Manager, if you accidentally generate business entity schema for Customer 360, the Configuration Manager overwrites your business entity configuration. After you generate the business entity schema, Customer 360 does not function as expected. For example, the task inbox and the Task Manager does not display any tasks even though you have valid tasks.

To resolve this issue, perform the following tasks:

1. In the Configuration workbench of the Hub Console, click **Repository Manager**.

   The Repository Manager tool opens.
2. To select the source repository, perform the following tasks:
   a. On the **Promote** tab, click **Visual**.
   b. Click the **Select** button next to the **Source** list.

      The **Open Repository** dialog box appears.
   c. On the **File Repository** tab, browse to the following directory:
      - For Party table-based data model. `<MDM installation directory>/app/tcr/hub/change-xml`
      - For Party Role table-based data model.`<MDM installation directory>/app/tcr/hub/change-xml/pre_c360_10.2_hf2`
   d. Select the `TCR_HUB.change.xml` file, and click **Open**.

      The Repository Manager tool loads and validates the repository from the selected file.
   e. Click **OK**.

      The Repository Manager tool displays the source repository as a hierarchical tree.
3. To select the target repository, in the **Target** list, select the ORS of Customer 360.
4. In the hierarchy tree of the source repository, select **Business Entity/Business Entity Services Configurations**.
5. To promote the selected object, perform one of the following tasks:
   - Right-click the selected object, and select **Promote**.
   - Click the **Promote** icon.
   - Drag the selected object from the source repository to the target repository.

   The **Promotion** dialog box appears.
6. Select **Replace the target objects with source objects**, and click **OK**.

   The **Impact Analyser** dialog box appears.
7. Click **OK**.
8. Click the **Apply Changes** icon.
9. If you have extended your business entities, restore or reconfigure the extended objects.
10. On the **Validate** tab, select the ORS of Customer 360, and click **Validate**.
11. Reopen a browser, and log in to Customer 360.

Import of the IDD ZIP file fails in IBM DB2.

In a WebSphere and IBM DB2 environment, when you import an IDD application in the Zip format, the import might fail.

To resolve the issue, use one of the following options:

- Increase the size of BLOB_DATA column to 30 MB in the CMX_SYSTEM.C_REPOS_DS_CONFIG table with the following IBM DB2 command:

  ```
  ALTER TABLE CMX_SYSTEM.C_REPOS_DS_CONFIG ALTER COLUMN BLOB_DATA SET DATA TYPE BLOB(31457280);
  ```
- Import and deploy the `BDDConfig.xml` file.

The Provisioning tool does not respond.

In the `C_REPOS_CO_CS_CONFIG` repository table, if the `TASKS` configuration contains the `LookupSegmentValue` business entity for a trigger, the Provisioning tool does not respond.

To fix this issue, remove the `LookupSegmentValue` business entity from the `TASKS` configuration. However, after you remove the `LookupSegmentValue` business entity from the `TASKS` configuration, you cannot add a new tag to a business entity.

Creating a record in the Customer 360 portal fails when Address Validation is turned off.

You cannot create a record in the Customer 360 portal with address validation turned off.

When you create a record, ensure the **Validate Address** option is turned on. The address validation process validates the addresses and populates the attributes, such as city and pincode.

If you don't want to use address validation in the customer 360 portal, perform the following steps:

1. Log in to the Hub Console. Before you make a change, acquire a write lock.
2. Under the **Model** workbench, click **Cleanse Functions**.
3. Expand **Portal Validations** and select **Address Verification**.
4. Click the **Details** tab.
5. In the **Details** tab, update the **Validate** field value from 1 to 0.
6. Click **Save**.

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